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Deal or no deal, investors are both concerned and excited

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Authors

Luke Templeman
Research Analyst

Galina Pozdnyakova
Research Analyst

Executive summary

Five months into the US-Iran conflict and the strength of corporate earnings makes it seem like the optimists have won. But things are more complicated. This report analyses proprietary data from our dbDataInsights team and shows that the issue of investing against the backdrop of a volatile energy market is highly polarising for people. Indeed, of all the investment themes we track, energy has one of the highest rates of both optimism and pessimism.

So, we have an odd situation: On one hand, global corporate earnings are incredibly strong despite the energy shock this year. On the other hand, the Iran conflict is still a massive problem for the world economy. An amicable agreement thus remains elusive.

Three findings from our analysis stand out.

— Energy – whether measured through prices or transition – is now one of the most-cited investment themes for US investors behind AI, and is prompting portfolio changes on par with technology and geopolitics.
— Energy prices carry one of the highest perceived risks of any theme we tested among UK investors.
— Our megatrend model shows that energy disruption (as a medium-term trend) has remained a positive force for the world economy this year. That pattern closely echoes the 1970s oil shocks.

Energy shocks are unambiguously bad for consumers and growth in the short term. But over longer time frames, our model shows that energy disruption has tended to be a net positive for markets and economies because it accelerates diversification of supply and pushes economies to generate more GDP per unit of energy consumed. We see early evidence of a similar dynamic beginning today.

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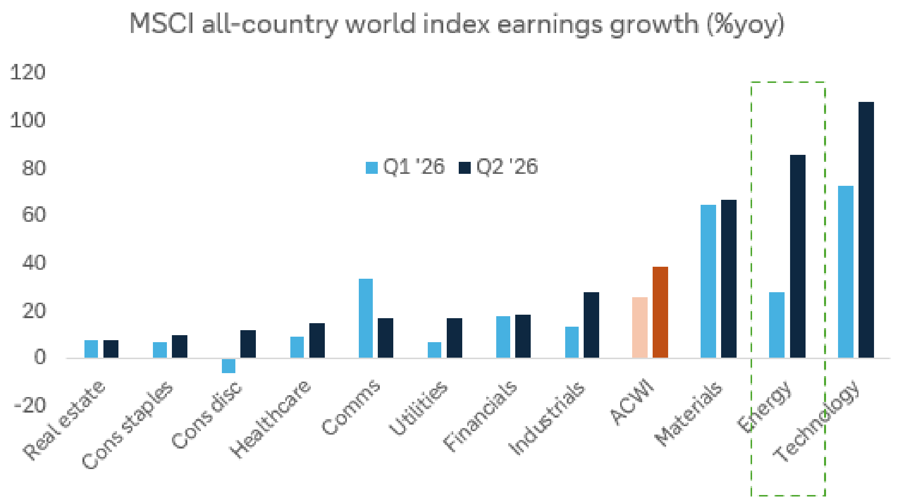
A serious issue ... but surprising resilience

The closure of the Strait of Hormuz has not been the economic catastrophe that many expected five months ago. In fact, corporates earnings have just had one of their best quarters outside of recession recoveries. For the companies in the MSCI all-country world index, median earnings growth is running at 16.5% yoy in Q2, well above the 14.7% in Q1. Energy has been a big part of that, with earnings growth, at a sector level, surging 86% in Q2, about triple the growth recorded in the prior quarter.

Figure 1: Earnings growth has been very strong in Q2



Source: Bloomberg Finance L.P., Deutsche Bank Asset Allocation

Economies have fared less well than companies, but they remain remarkably resilient given the harsh predictions that came from some quarters in early March. Of course, stockpile releases and other buffers have helped smooth the disruption – and as these wear down further, larger problems may arise.

So, now we have a curious juxtaposition of outlooks. On one hand, corporate earnings are proving resilient and the long-term focus on energy security has helped make many of the associated equities good investments. On the other hand, the Iran conflict is still a massive problem for the world economy.

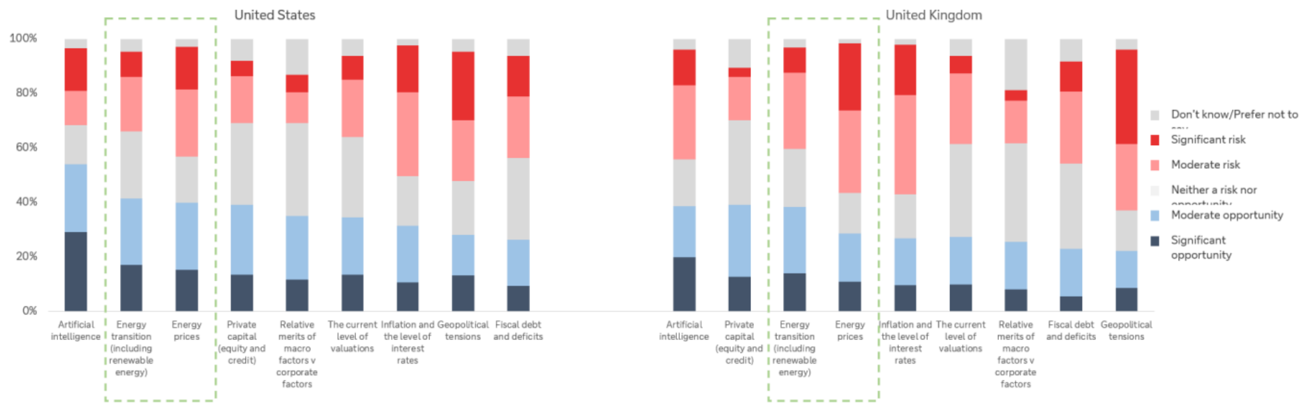
What investors think about the effect of energy on their portfolio

Our dbDataInsights survey on the topic was carried out in April and May of this year and polled a representative sample of people in the US and the UK. The responses indicate that energy has become one of the two dominant investment themes of 2026 – alongside, and closely intertwined with, artificial intelligence.

Energy is a high-conviction theme – in both directions

When we asked investors whether they see various investment themes as a risk or an opportunity for 2026, energy transition and energy prices ranked just behind artificial intelligence as the most-cited opportunities in both the US and the UK. The below chart shows a selection of the themes.

Figure 2: Survey question: "For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



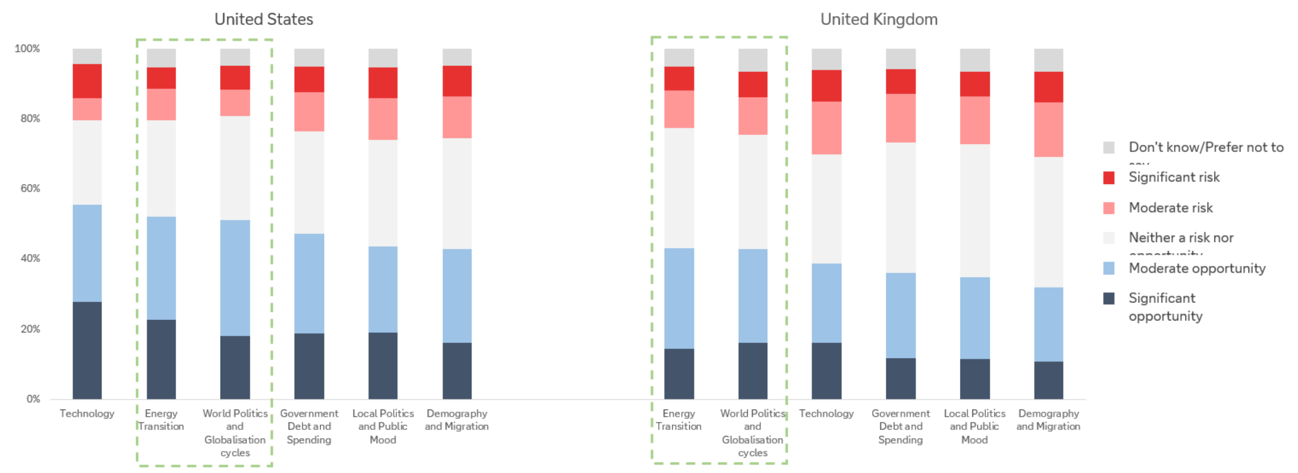
Source: dbDataInsights, Deutsche Bank

Energy is already changing portfolios

Conviction about energy appears to be translating into action – particularly when we think about it from a medium- or long-term point of view. To assess this, we fell back on our megatrend model that assesses the development of the six key megatrends that are driving the world's economies and markets.

When we asked investors how likely they are to change their portfolio over the next 12 months because of each megatrend, a dominant theme was energy and geopolitics as the below chart shows.

Figure 3: Survey question: “Within the next 12 months, how likely are you to make changes to your investment portfolio due to the following thematic megatrends?”

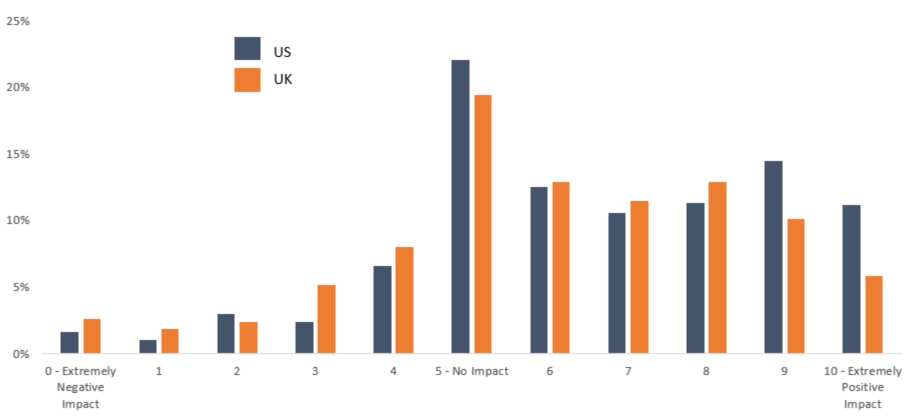


Source: dbDataInsights, Deutsche Bank

Taken together, the two survey questions tell a consistent story: energy has moved from a background, structural megatrend to one that investors are actively trading around, much as technology has been for the AI boom. The difference is that, unlike technology, energy's near-term path is being set as much by missile strikes and shipping insurance premia as by fundamentals.

Investors appear to sense this: when we asked how the energy transition will affect their main portfolio over the next three years, responses skewed firmly positive.

Figure 4: Investors’ anticipated impact of the energy transition on their main investment portfolio over the next 3 years

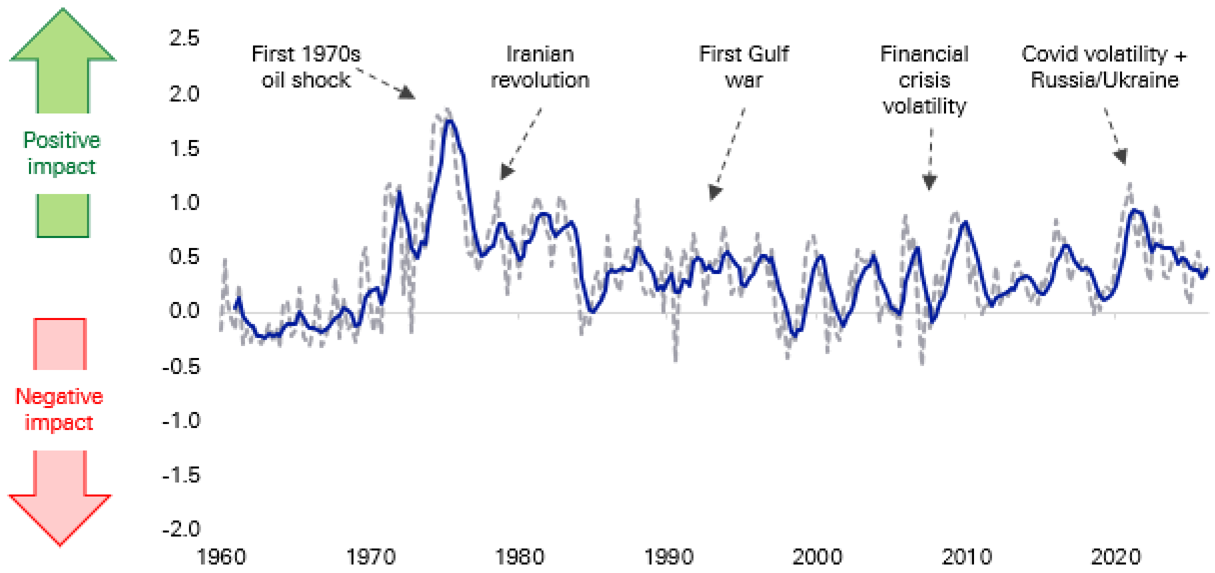


Source: dbDataInsights

Energy as a megatrend: why the medium-term signal is so positive

Energy is one of the six long-term trends that feeds into our megatrend model that uses almost 100 data series since 1957 to track their impact on economies and markets. Energy is a nuanced megatrend as, unlike several others, there is a big difference between the effect of short-term disruption and long-term gain. Indeed, prior periods of energy disruption have been incredibly good for the long-term energy trend, even if the short-term disruption was incredibly painful for consumers and economies.

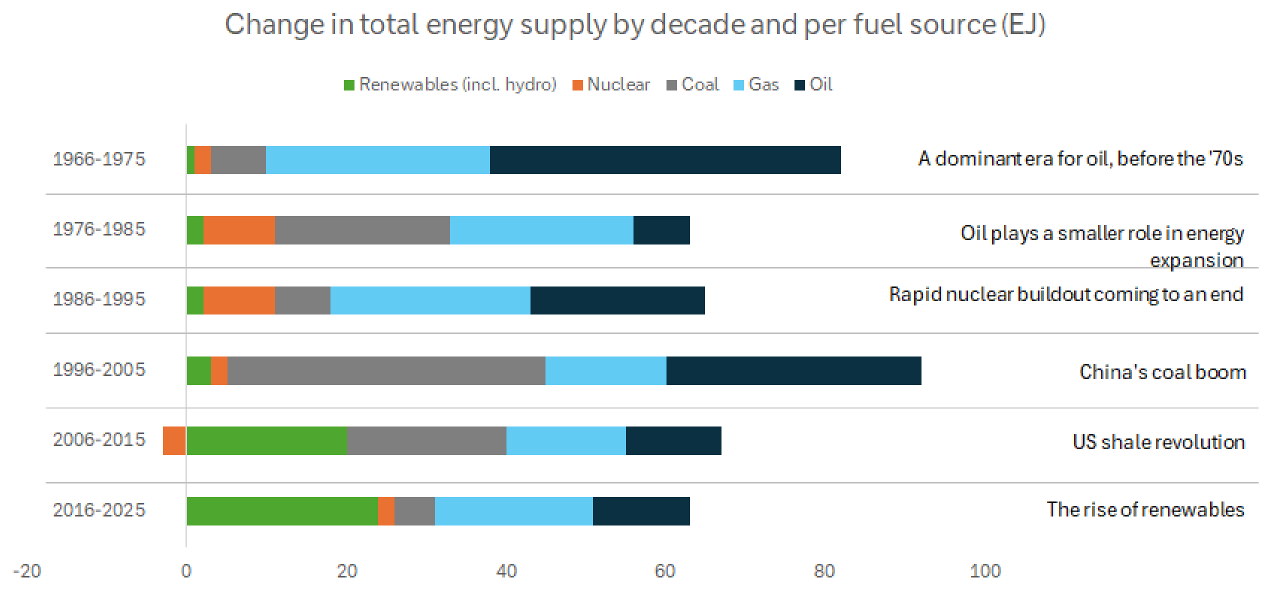
Figure 5: Our energy megatrend indicator has generally been positive as the world's economies diversify their supplies at the same time that innovation makes the economy more energy efficient



Source: Factset, Bloomberg Finance L.P., Fineon, EIA, Deutsche Bank Research

Viewed through a medium-term lens, the last five months of conflict in the Middle East can be viewed as likely being a long-term positive for economies and markets. That is because the Iran war is accelerating exactly the kind of diversification our model rewards: European and Asian buyers have moved further into US, Guyanese and West African crude and LNG since the Strait's closure; several G20 governments have fast-tracked strategic reserve releases and non-Gulf supply agreements; and renewed urgency has attached to the 'friend-sufficient' push for critical minerals and energy inputs that began during the US-China strategic rivalry.

Figure 6: Shocks over the decades have defined several eras for the evolution of the energy mix



Source: Energy Institute

The 1970s is something of a parallel

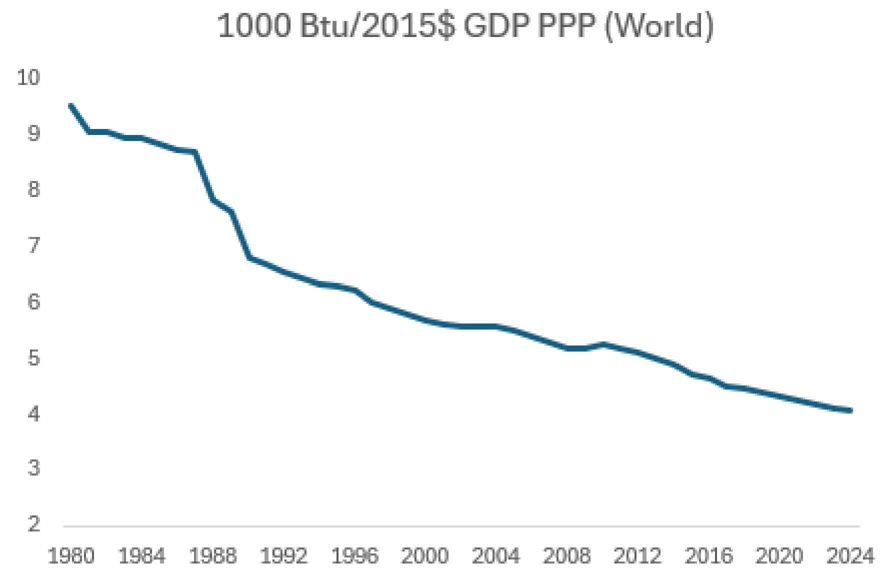
The closure of the Strait of Hormuz in 2026 and the disruption to Russian gas flows into Europe in 2022 both echo the twin oil shocks of 1973 and 1979, when OPEC's move to set its own prices followed by the Iranian revolution, sent crude to ten times its early-1970s level and permanently altered the trajectory of global oil demand.

Global oil consumption growth slowed from 7.8% before 1973 to 2.2% pa for most of the rest of the decade (on calculations from the Energy Institute). It outright declined after the second and took a decade for global oil consumption to recover to its 1979 level.

Whilst the 1970s shocks created havoc for both individuals and economies, they accelerated the pace at which the world changed how it generated economic output. The energy intensity of GDP fell 3.3% in 1980 (the largest single-year improvement in the Energy Institute's data series). That adjustment reflected both behavioural change and, over the following decade, structural investment in fuel efficiency and the phase-out of oil-fired power generation.

Since 1980, the energy intensity of GDP has roughly halved as shown in the following chart.

Figure 7: The energy intensity of GDP has been steadily falling – partly because of efficiency and innovation



Source: EIA

This trend is captured in our megatrend model which shows the medium-term payoff from decades of energy disruption. Those forces continue today. Every episode of chokepoint risk, from Ukraine in 2022 to Hormuz in 2026, adds another increment to the diversification of global energy supply and another incentive for energy innovation across economies.

Conclusion

Five months into the US-Iran conflict war, investors are highly polarised about the investment merits of investing against the backdrop of energy volatility. However, those that are positive are very positive. Our megatrend model suggests that reaction is rational: like the 1970s oil shocks, the medium-term legacy of this conflict is likely to be faster energy diversification and a further, durable improvement in the energy intensity of GDP, even though the near-term path for prices is likely to stay volatile.

As a result, energy security and supply diversification should keep attracting capital regardless of where the oil price settles. The beneficiaries are likely to cascade up and down the value chain including, LNG infrastructure, non-Gulf production, renewables, strategic reserves and grid resilience as countries and companies all try to diversify their chokepoint risk.

Appendix 1

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Korea: From Industrialization to Internationalization

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Authors

Perry Kojodjojo
Strategist

Chen Kan
Macro Strategist

Executive Summary

Korea sits at the intersection of some of the most important structural forces reshaping the global economy. Re-industrialization, AI infrastructure investment and supply-chain diversification are directing capital toward industries where Korea already possesses scale, technological leadership and established industrial ecosystems. At the same time, policymakers are reinforcing these advantages by supporting strategic industries and expanding Korea's industrial footprint globally.

Yet a paradox remains. Korea's economic influence has expanded much faster than its financial influence. Despite accounting for roughly 2% of global GDP and playing a central role in several strategic industries, the Won remains only a modestly used international currency and Korean assets continue to trade at valuation discounts. Korea's internationalization agenda seeks to narrow this gap, building a financial system that better reflects the country's growing importance in global trade, manufacturing and technology.

In this report, we argue that:

— **The world is increasingly investing in industries where Korea already leads,** as re-industrialization, AI investment and supply-chain diversification redirect capital toward Korea's areas of strength.

— **Korea already possesses what others are trying to build,** with its strengths in manufacturing, technology, innovation and competitiveness.

— **Internationalization is the next stage of development,** helping Korea retain and intermediate a larger share of the wealth it creates.

— **Korea is pursuing a distinct and phased model of internationalization,** combining market-access reforms, greater offshore usability and gradual liberalization while preserving financial stability.

— **The key market implication is a gradual narrowing of the gap between Korea's economic footprint and financial footprint,** as stronger industrial performance increasingly translates into deeper capital markets, a more internationally used currency and stronger demand for Korean assets.

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The world is investing in the industries where Korea already leads

Industrial policy has returned to the centre of global economic strategy. Across the US, Europe, Japan, China and much of Asia, governments are deploying subsidies, tax incentives, localisation requirements and strategic investment programmes to strengthen domestic manufacturing, secure access to critical technologies and reduce supply-chain vulnerabilities. While the policies differ across jurisdictions, the direction of travel is increasingly aligned - resilience, technological leadership and economic security now rank alongside efficiency as core policy objectives.

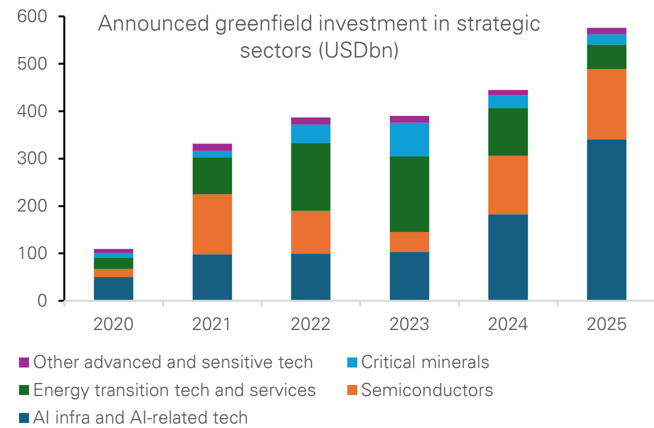
Figure 1: Industrial policy comparison — At-a-Glance

	Policy direction	Strategic sectors	Main objective	Incentive model
Advanced economies				
US	Reshore critical technology production	Chips; EVs/batteries; clean energy; defence; critical minerals	Secure strategic supply chains with trusted partners	Subsidies, tax credits, grants, loans, export controls
EU	Make decarbonisation an industrial growth engine	Net-zero industry; clean tech; heavy industry; raw materials	Build competitive clean industry while protecting competitiveness	State-aid flexibility, EU funding, carbon rules, faster permits
Japan	Strengthen economic security and low-carbon industry	Chips; batteries/EVs; defence; aerospace; nuclear/clean energy	Secure advanced inputs and reinforce defence/energy resilience	Targeted subsidies, tax breaks, procurement, energy investment
North Asia				
China	Pursue self-sufficiency and technology leadership	Chips; EVs/batteries; shipbuilding; green tech; critical minerals	Reduce external vulnerability while preserving export strength	Subsidies, cheap credit, tax relief, procurement, export controls
Korea	Scale national champions in strategic technologies	Memory chips; AI; EV batteries; autos; shipbuilding; defence; nuclear	Hold top-tier chip position and diversify growth engines	Mega capex funds, policy credit, tax credits, R&D and talent programmes
Taiwan	Extend semiconductor leadership into AI hardware	IC design/manufacturing; AI servers; smart manufacturing; digital sectors	Use AI + chips to deepen the innovation ecosystem	R&D credits, shared labs, pilot lines, talent/startup support
South Asia				
India	Scale manufacturing and domestic value-add	PLI sectors; electronics; autos/EVs; chips; pharma; capital goods	Become a high-value manufacturing and export hub	Output-linked incentives, sector schemes, infrastructure/logistics funding
Indonesia	Move nickel downstream into EV value chains	Nickel; battery materials; cells; EV assembly; clean-energy metals	Shift from raw materials to integrated battery/EV production	Export bans, smelter controls, industrial parks, state-firm ecosystems
Malaysia	Upgrade from assembly to higher-value manufacturing	E&E; semiconductor back-end; advanced materials; aerospace; pharma	Move up technology and Industry 4.0 capability	Tax incentives, grants, roadmaps, cluster facilitation
Philippines	Revive manufacturing around electronics and components	Electronics packaging; autos; aerospace components; broader manufacturing	Deepen regional supply-chain participation beyond services	SEZ incentives, FDI promotion, sector roadmaps, streamlined approvals
Singapore	Anchor high-value advanced manufacturing	Precision engineering; chips; pharma; robotics; automation; digital manufacturing	Keep high-value production linked to R&D and HQ functions	Grants, co-funding, R&D incentives, skills schemes, bespoke EDB packages
Thailand	Use EEC and S-curves to upgrade industry	EVs; smart electronics; aviation/logistics; robotics; bio/medical industries	Become a regional hub for upgraded manufacturing	SEZ tax breaks, infrastructure, premium sector incentives
Vietnam	Build supporting industries and semiconductor capability	Electronics support; mechanical/autos; textiles/footwear; high-tech; chips	Raise localisation and integrate deeper into global supply chains	Cost-sharing subsidies for R&D, technology and workforce spending

Source : Deutsche Bank Research

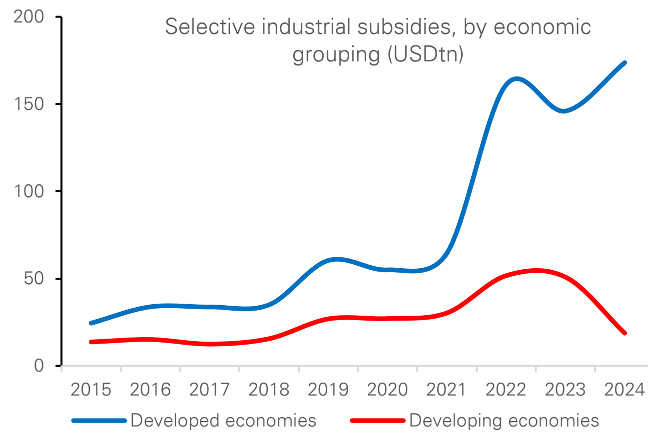
The scale of this shift is already visible in global investment flows. According to UNCTAD, strategic sectors—including semiconductors, clean energy, advanced manufacturing, defence and digital infrastructure—accounted for 44% of global greenfield investment in 2025, up from just 16% in 2020. Over the same period, the value of announced projects rose from $109bn to $576bn, highlighting the growing concentration of capital in strategic industries. This is creating one of the largest industrial investment cycles in decades and directing investment towards precisely the sectors where Korea has already established globally competitive capabilities.

Figure 2: Growing investments in strategic sector globally



Source : Deutsche Bank Research, UN Trade and Development (UNCTAD). Note: Data for 2025 are annualized on the basis of information available as of 30 November.

Figure 3: Increase in the use of subsidies to support growth in the strategic sector



Source : Deutsche Bank Research, UN Trade and Development (UNCTAD), based on New Industrial Policy Observatory (NIPO). Note: Only subsidies worth more than $10mn are recorded in the NIPO database.

Three structural forces are now converging in Korea's favour.

- **Reshoring, ally-shoring and rearmament are increasing global demand for industries in which Korea already holds a competitive advantage.** Semiconductors, shipbuilding, defence equipment, batteries and critical minerals have become central to national-security and supply-chain strategies. Post-Ukraine European rearmament has increased demand for reliable defence suppliers, while the US and its allies are seeking trusted partners to rebuild strategic supply chains. Korea already possesses many of the industrial capabilities that advanced economies are now attempting to reshore or source from aligned partners.
- **The global AI capex supercycle is providing a major terms-of-trade uplift**. Hyperscaler capital expenditure is estimated to be growing at close to 70% YoY through 2024-26, creating strong demand for GPUs, HBM, DRAM and related semiconductor infrastructure. As large language models become increasingly complex and agentic AI requires greater processing power and speed, demand for advanced memory chips is rising sharply. Samsung Electronics and SK Hynix place Korea at the centre of this cycle, positioning the country as one of the most direct beneficiaries of the global AI infrastructure buildout.
- **China de-risking is reshaping supply chains in Korea's favour.** Western buyers are actively reducing reliance on Chinese supply chains across selected strategic sectors, including batteries, battery materials, vessels, robotics and critical minerals. Korea is one of the few economies with both the industrial scale and geopolitical alignment to absorb some of the production, investment and market share being redirected away from China. While Chinese overcapacity remains a competitive challenge in some industries, the global effort to diversify away from China is simultaneously creating opportunities in strategic sectors where technology, reliability and security considerations matter more than cost alone.

This convergence matters. Korea is not benefiting from one isolated export cycle. **Rather, it sits at the intersection of ally-shoring, AI infrastructure investment and China supply-chain diversification—three structural forces that are likely to persist for years.** In many respects, the world is increasingly investing in precisely the industries where Korea already possesses scale, technological leadership and established industrial ecosystems.

Figure 4: Korea ranks among the world's best positioned economies for re-industrialisation

Rankings (1 = Best)													
Metric	US	EU	Japan	China	Korea	Taiwan	Singapore	Malaysia	Indonesia	Thailand	Vietnam	Philippines	India
Manufacturing value added (% of GDP)	13	11	8	3	2	1	9	6	7	5	4	10	12
Share of high-tech exports	11	10	7	8	2	1	3	6	13	9	5	4	12
Current account balance (% of GDP)	13	8	5	6	4	1	2	9	10	7	3	12	11
Government debt (% of GDP)	12	10	13	11	5	2		8	4	7	3	6	9
Gross capital formation (% of GDP)	11	13	6	1	5	7	10	9	3	12	4	8	2
Total surplus capacity (MW)	3	2	5	1	6	10	13	12	7	9	8	11	4
Electricity price (USD/kWh)	8	12	11	4	5	10	13	7	1	6	2	9	3
Business infrastructure	1	7	6	3	5	4	2	8	12	9	10	13	11
Global talent competitiveness index	3	4	5	8	6	2	1	7	12	11	10	9	13
Regulatory quality	2	5	3	11	6	4	1	7	8	9	13	10	12
Trade freedom	12	6	8	9	10	2	1	3	7	11	5	3	13
Broad competitiveness	3	10	9	4	6	2	1	5	13	7	8	12	11
Economic Complexity	6	7	1	5	3	2	4	8	13	9	12	11	10

Source :Deutsche Bank Research, CEIC, Haver Analytics, Insead, World Bank, The Heritage Foundation, IMD.org, Harvard Business School

Policy is also reinforcing these advantages. Rather than creating entirely new industries, Korean policymakers are largely focused on scaling existing strengths, securing critical inputs and supporting the global expansion of Korean industrial ecosystems. Recent initiatives have directed capital toward semiconductors, AI infrastructure, batteries, defence and other strategic sectors through policy financing, investment programs and regulatory support. At the same time, Korea is strengthening access to critical minerals, expanding overseas industrial partnerships and facilitating outward investment by Korean firms. The objective is not simply to support national champions, but to reinforce the broader ecosystem around them and ensure Korea remains embedded in the next phase of global industrial development.

Korea already possesses what others are trying to build

Few economies are better positioned to benefit from the new industrial order than Korea. Unlike many countries attempting to rebuild manufacturing capacity from a relatively low base, Korea already has one of the world's most advanced industrial ecosystems.

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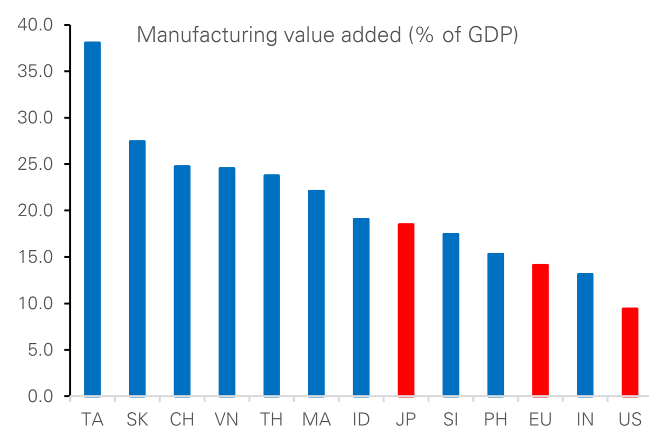
- **Industrial ecosystems matter.** Korea has built globally competitive positions across semiconductors, batteries, shipbuilding, defence, robotics, nuclear energy and advanced manufacturing. These industries are supported by a deep industrial ecosystem spanning engineering talent, R&D capabilities, specialised supplier networks and world-class infrastructure and relatively healthy public and external balances. These capabilities are difficult to replicate quickly and increasingly valuable in a world prioritising resilient and secure supply chains.

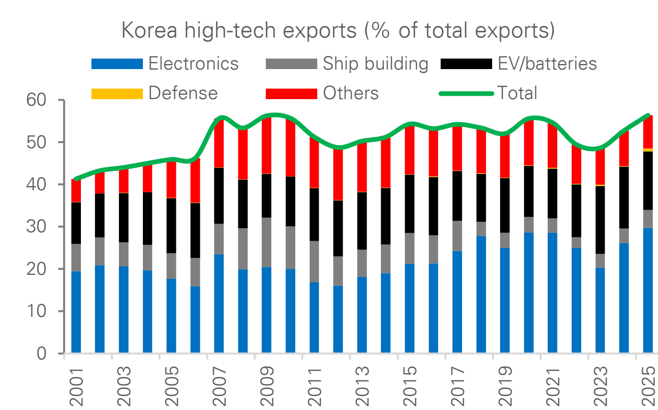
- **Starting from strength matters.** Manufacturing accounts for approximately 24.3% of GDP, among the highest shares in the OECD and roughly two times the level of the advanced economies. At a time when governments are seeking to expand domestic production of strategic goods, Korea enters the re-industrialisation cycle with an existing manufacturing base rather than having to build one from scratch.
- **Technology leadership matters.** Approximately 58% of Korea's exports are high-tech products, one of the highest shares among major economies, while semiconductors account for roughly 50% of high-tech exports. Korea also accounts for ~60% of global memory semiconductor production and 90% of global HBM supply, making it one of the most important beneficiaries of the AI infrastructure cycle.

Figure 5: Manufacturing accounts for approximately 24.3% of GDP, among one of the highest shares in the OECD and roughly two times the level of the advanced economies.



Source : Deutsche Bank Research, Haver Analytics

Figure 6: Korea's export engine increasingly driven by High-Tech industries

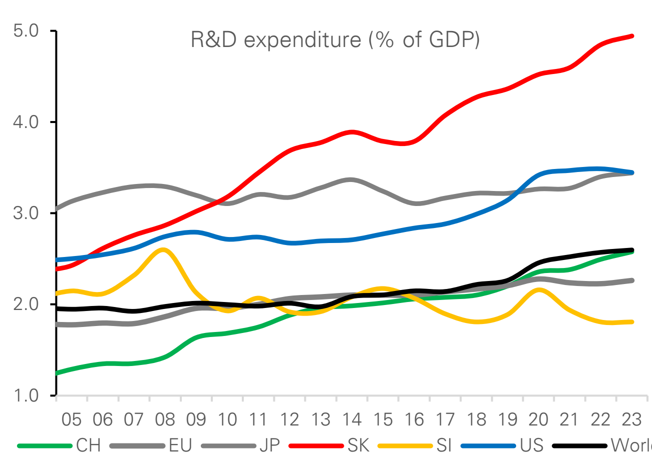


Source :Deutsche Bank Research, ITC

Figure 7: Significant improvement in terms of trade since 2024 for Korea

Source : Deutsche Bank Research, Bloomberg Finance LP, Haver Analytics

Figure 8: Korea has one of the highest R&D expenditures in the world



Source : Deutsche Bank Research, World Bank

- **Strategic industries matter**. Beyond semiconductors, Korea has built globally competitive positions in EV batteries, defence, advanced shipbuilding, robotics and nuclear energy. Korean firms account for ~30% of global EV battery capacity and 90% of global LNG carrier orders, while defence exports are increasingly benefiting from rising global security spending and rearmament efforts.
- **Energy increasingly matters.** Under South Korea's latest electricity plan, nuclear power is expected to account for approximately 35% of electricity generation by 2038, while carbon-free sources are projected to supply roughly 70% of total generation. The plan is designed in part to accommodate rising electricity demand from semiconductors, AI data centres and broader industrial electrification. As manufacturing and digital infrastructure become increasingly energy-intensive, access to reliable, scalable and lower-carbon power is emerging as an important component of industrial competitiveness.

Korea's advantages extend beyond individual industries. Through leadership in semiconductors, batteries, shipbuilding, defence and advanced manufacturing, Korea is becoming an increasingly important industrial partner for both advanced and emerging economies. As supply chains become more closely linked to economic security, reinforcing its role within the evolving architecture of the global economy. Rather than acting solely as an exporter of goods, Korea is increasingly participating in the development and expansion of industrial ecosystems across multiple regions through investment, technology transfer and supply-chain partnerships. In that sense, **Korea's industrial strengths are not only supporting growth and competitiveness at home but are also reinforcing its role within the evolving architecture of the global economy.**

Korea's industrial position remains strong, but its advantages should not be taken for granted. Despite leadership in semiconductors, shipbuilding, defence and advanced manufacturing, Korea remains dependent on imported energy, critical minerals and raw materials, leaving parts of its industrial ecosystem exposed to commodity and supply-chain shocks. Competition is also intensifying. China continues to move up the value chain across batteries, shipbuilding, robotics and segments of the semiconductor supply chain, while governments globally are deploying increasingly aggressive industrial policies to build domestic capabilities. Industrial leadership is rarely permanent. **Korea's re-industrialisation story therefore depends not only on existing strengths, but also on its ability to sustain innovation, secure critical inputs and preserve its technological edge over the coming decade.**

Internationalization as the next stage of development

Korea scores strongly on measures of industrial depth, export sophistication and external strength. Manufacturing value added, high-tech exports and current-account balances compare favourably with most regional peers. Yet Korea remains among the cheapest major Asian markets on P/E and P/B metrics, while the Won screens as undervalued on our valuation framework.

Figure 9: Korea's valuation paradox: strong fundamentals, cheap assets

	Economic Matrix			Financial Market Matrix				
	Manufacturing value added (% of GDP)	Share of high-tech exports (%)	Current account bal (% of GDP)	12M forward P/E ratio	ROE (%)	12M forward P/B ratio	Dividend yield	FX valuation
US	9	28	-4	21.3	20.0	4.6	1.2	5.7
EU	14	31	2	15.8	12.6	2.2	3.1	-1.6
Japan	18	40	5	22.4	12.0	2.7	1.5	-24.7
China	25	35	4	14.9	9.9	1.6	2.6	-5.3
Korea	27	59	7	6.9	10.5	1.5	1.9	-15.0
Taiwan	38	78	17	20.3	13.9	4.0	2.1	-5.3
Singapore	17	55	17	17.2	9.9	1.8	4.1	4.9
Malaysia	22	48	2	15.1	10.6	1.5	4.3	1.8
Indonesia	19	9	0	10.0	11.8	1.3	6.3	-7.8
Thailand	24	31	3	16.1	8.6	1.6	3.6	7.5
Vietnam	25	48	7	12.0	15.1	1.7	1.9	
Philippines	15	53	-3	9.9	11.6	1.2	3.7	2.9
India	13	23	-1	19.9	14.2	2.6	1.8	-6.6

Note: FX valuation is based on the simple average of three valuation models: FEER, DBEER, and PPP. Positive (negative) values indicate overvaluation (undervaluation).

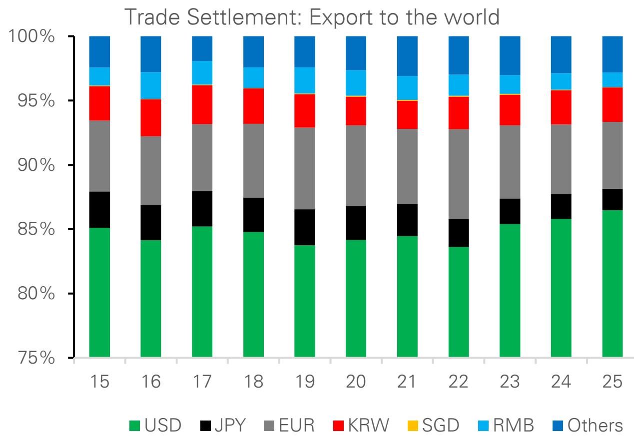
Source :Deutsche Bank Research, CEIC, Haver Analytics, Bloomberg Finance LP. Note: FX valuation is based on the simple average of three valuation models: FEER, DBEER, and PPP. Positive (negative) values indicate overvaluation (undervaluation).

Our preferred valuation framework suggests the Won remains materially undervalued relative to fundamentals. Korea's current account surplus could reach ~10.3% of GDP this year and remain above 5% over the medium term, well above its long-run average of around 3.5% of GDP. Under a FEER framework, which anchors exchange rates to external-balance sustainability, such an external position is inconsistent with current KRW levels and **implies that the Won is undervalued by ~ 10% in trade-weighted terms.**

More importantly, this result may prove conservative. Korea's current account improvement increasingly reflects a positive terms-of-trade shock driven by AI-related semiconductor demand, rising export sophistication and stronger pricing power. As a result, Korea's industrial renaissance may not only justify larger external surpluses, but also a higher equilibrium valuation for the Won.

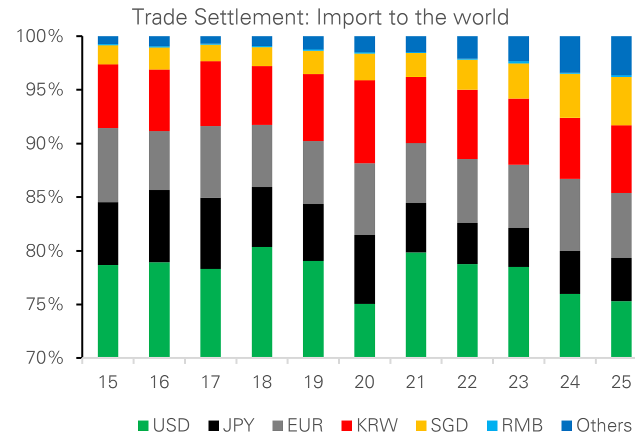
Besides asset valuation discounts, Korea's growing economic importance is not reflected in its financial system and currency yet. Today, Korea accounts for roughly 2% of global GDP, ranks among the world's largest exporters of advanced manufactured goods and is home to several globally strategic industries. Won accounts for approximately 0.9% of global FX turnover, which puts it at around 12th rank; while its role in international payments, funding markets and reserve holdings remains considerably smaller than Korea's relative economic quantum.

Figure 10: Korea's exports are still overwhelmingly settled in foreign currencies



Source : Deutsche Bank Research, Haver Analytics

Figure 11: The won plays only a limited role in Korea's import settlement

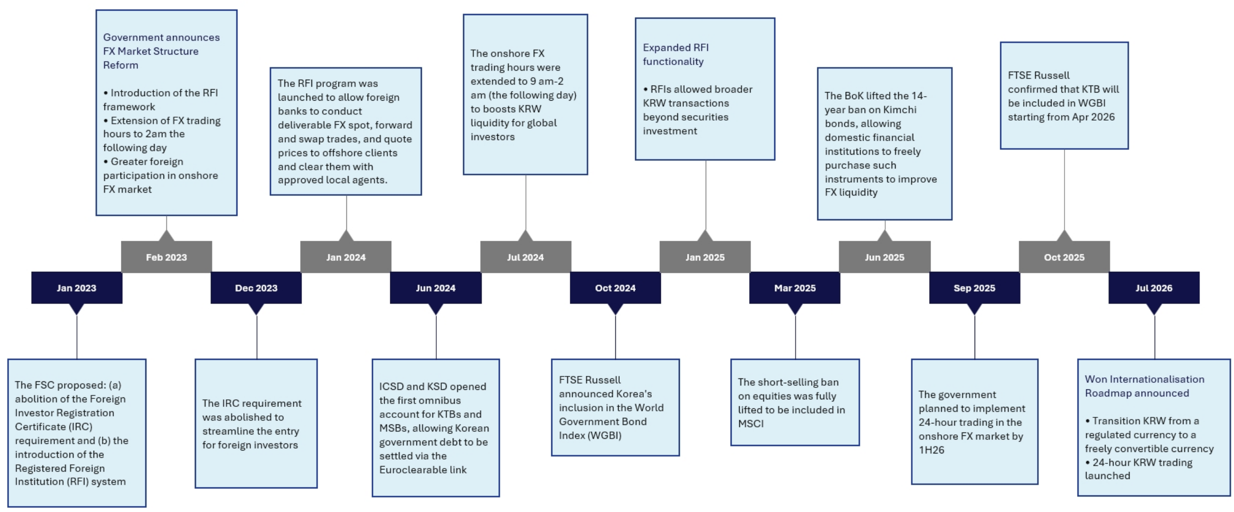


Source : Deutsche Bank Research, Haver Analytics

This gap between economic importance and financial influence helps explain the government's newly announced roadmap for Won internationalization. The objective is not to transform the Won into an alternative reserve currency, nor to replicate China's offshore RMB model. Rather, policymakers are focused on creating the conditions for a more internationally usable currency — one that is easier to trade, fund, settle and hold across time zones. The roadmap combines the development of offshore settlement and funding infrastructure with a gradual easing of selected capital-account restrictions, allowing international use of the Won to expand while maintaining a measured and risk-managed approach to liberalization.

More fundamentally, Korea's push to internationalize the Won reflects a broader question about how the country captures the benefits of its industrial success. Our work argued that re-industrialization, the AI infrastructure cycle and supply-chain diversification are likely to strengthen Korea's external balances and generate larger pools of national savings. Yet generating wealth and retaining wealth are not the same thing. A significant share of the savings generated by Korea's industrial success continues to be recycled abroad through portfolio investment and overseas asset accumulation. **Strong external balances therefore do not automatically translate into deeper domestic capital markets, stronger asset valuations or a more internationally important currency.**

Figure 12: The reforms started well before July 2026



Source : Deutsche Bank Research

In many respects, Won internationalization can be viewed as the financial counterpart to Korea's industrialization. Industrialization generates exports, profits, savings and external surpluses; internationalization helps deepen capital markets, broaden participation and increase the likelihood that a greater share of the resulting trade, investment and funding activity is intermediated through Korean financial markets. **The objective is therefore not simply a more international currency, but a financial system that more closely reflects Korea's growing role in the global economy.** The result is not a wholesale opening of the capital account, but a gradual and carefully sequenced effort that combines greater offshore usability of the Won with the selective liberalization of domestic financial markets, while preserving financial stability and policy flexibility.

Korea's model – A phased approach to internationalization

A common mistake is to analyze Korea's internationalization strategy primarily through the lens of either China or Singapore. In reality, Korea is pursuing a distinct path that differs from both models.

China's approach relied heavily on the creation of a parallel offshore ecosystem that allowed the renminbi to internationalize while maintaining extensive control over domestic financial markets. Singapore represents the opposite approach, combining deep global integration with virtually unrestricted capital mobility. Korea's strategy sits somewhere between these two models. Policymakers are seeking to expand the usability of the Won while preserving financial stability and policy flexibility.

The first phase of reform focused on improving market accessibility. Measures such as the Registered Foreign Institution framework, omnibus accounts, extended FX trading hours and broader market-access reforms were designed to reduce operational barriers facing global investors and improve access to Korean financial assets. These reforms laid the foundations for greater foreign participation in Korean bond and equity markets.

The next phase goes further by focusing on the usability of the currency itself. Recent reforms are designed to make the Won easier to hold, transfer, fund and settle internationally. Rather than concentrating solely on attracting investors into Korean assets, policymakers are now seeking to expand the range of activities that can be conducted in Won across investment, funding and settlement activities.

Importantly, liberalization is being implemented gradually. Korea is not pursuing a wholesale opening of the capital account. Instead, reforms are being sequenced carefully alongside monitoring systems, liquidity facilities and macroprudential safeguards. The objective is to reduce frictions and improve accessibility without compromising financial stability.

The result is best understood as a phased transition from market-access reform towards broader currency internationalization. Earlier reforms focused on making Korean financial assets easier for global investors to access. The next phase is aimed at making the Won itself easier to use across investment, funding and settlement activities.

Figure 13: Korea is pursuing a distinct model of currency internationalization

Korea is pursuing a distinct model of currency internationalisation				
	Singapore	**China**	**India**	**Korea**
Primary objective	Global financial centre	Global reserve currency	Expand INR use in trade and finance	Increase KRW investability and global usage
Internationalisation model	Full openness	Offshore-onshore framework (CNH/CNY)	Trade-led internationalisation	Phased internationalisation led by capital-market integration
Key policy tools	Open capital account, global financial intermediation	Offshore RMB markets, clearing banks, Connect programmes	INR trade settlement, bilateral arrangements	FX market liberalisation, benchmark inclusion, settlement infrastructure
Treatment of offshore markets	Integrated	Separate offshore ecosystem	Limited	Greater offshore usability through infrastructure linked to the domestic market
End-state vision	Fully open financial centre	Widely used international currency	Broader regional and trade usage	Globally investable and increasingly usable currency
Key challenge	Small domestic market	Capital controls	Limited global demand	Balancing openness and financial stability
Sequencing	Openness first	Trade settlement → Offshore ecosystem → Investment integration	Trade settlement → Gradual financial liberalisation	Market access → Broader currency usage

Source : Deutsche Bank Research

How Korea is building a more international Won?

The first pillar is market accessibility. Korea has already moved to 24-hour dollar-Won spot trading, allowing investors to trade and hedge Won exposure across Asian, European and North American trading sessions. Earlier reforms such as the Registered Foreign Institution (RFI) framework, omnibus accounts and settlement-system upgrades were designed to reduce many of the operational frictions that historically limited foreign participation in Korean financial markets. These measures laid the foundations for greater foreign ownership of Korean bonds, equities and other financial assets.

The second pillar is offshore usability. The proposed framework allows registered Offshore Won Settlement Institutions to facilitate Won transfers, funding and settlement between non-residents without requiring every transaction to pass through traditional domestic account structures. A new BoK offshore settlement network is intended to support 24-hour settlement and improve the ability of investors, corporates and financial institutions to hold, transfer and deploy Won outside Korean business hours. Authorities are also seeking to encourage greater use of deliverable KRW markets through offshore settlement infrastructure and measures that support deliverable rather than non-deliverable instruments over time. This reflects one of the central features of the roadmap: expanding the ability of non-residents to hold, transfer, fund and settle Won offshore, even as onshore liberalisation remains gradual and carefully sequenced.

The third pillar is gradual liberalization. Rather than removing all restrictions at once, policymakers are selectively easing reporting requirements, simplifying account structures and gradually shifting certain transactions from prior approval to ex-post reporting. Importantly, liberalization is occurring within a controlled framework: offshore Won transactions initially operate through registered institutions, while market monitoring, liquidity facilities and prudential safeguards are expanded in parallel.

Figure 14: The building blocks of KRW internationalization

Pillar	What has changed	Why it matters
Market accessibility	24-hour USD/KRW trading; RFI framework; omnibus accounts; settlement upgrades	Allows foreign investors to access, trade and hedge Korean assets more easily across global time zones
Capital market integration	WGBI inclusion, MSCI market-access reforms, Euroclear/ICSD connectivity, expanded collateral and repo infrastructure	Reduces operational barriers for global investors and integrates Korean assets more closely with global portfolios
Offshore won usability	Offshore Won Settlement Institutions and offshore settlement network	Allows non-residents to hold, transfer, fund and settle won outside Korea more efficiently
Regulatory liberalisation	Many inter-foreigner won capital transactions move toward lighter reporting and ex-post supervision	Reduces frictions for investors, corporates and treasury centres while maintaining oversight
Digital money infrastructure	Stablecoin framework, CBDC initiatives, tokenised government-bond pilots	Establishes future rails for cross-border payments and digital won usage

Source : Deutsche Bank Research

Taken together, these reforms represent an evolution from market-access reform towards broader currency internationalization. Earlier initiatives such as the RFI framework, omnibus accounts, extended trading hours and benchmark-related reforms focused primarily on reducing barriers for foreign investors. The new roadmap builds on that foundation by expanding offshore settlement infrastructure, easing selected capital-account restrictions and increasing the range of activities that can be conducted in Won. The objective is therefore not only to make Korean assets easier to access, but also to make the Won itself easier to hold, transfer, fund and settle internationally.

Market Implication: A new funding, investment and settlement currency for Asia?

For the currency, success should not be measured by whether the Won becomes a major reserve currency. **A more realistic benchmark is whether the Won gradually plays a larger role in economic activities linked to Korea's own industrial and trading footprint.** Progress would likely be visible through greater use of Won settlement in trade, broader participation by foreign investors in Korean financial markets, and increasing use of Won-denominated funding and hedging instruments. The reforms create the infrastructure necessary for such a transition, but market adoption will ultimately determine the outcome.

Figure 15: Measuring progress towards a more international Won

Success Metric	Korea	Japan	Australia	Why it matters
Foreign ownership of local-currency government bonds	28.7%	13.7%	52.6%	Higher foreign participation and broader investor base
Foreign ownership of domestic equities	38.5%	11.9%	30.9%	Increased representation in global portfolios
Institutional ownership of equities	20.0%	34.0%	60.0%	More stable portfolio flows
Inclusion in Global DM benchmark	Not in MSCI DM	Yes	Yes	Measures potential structural demand from its financial assets
Home-currency settlement share in external trade	3.0%	33% (exports only)	8.8%	Gradual increase in trade invoicing and settlement in won
Foreign participation in repo / funding markets	2.1%	N/A	60.0%	Deeper offshore funding and collateral markets
Currency share of global FX turnover	0.9%	8.5%	3.1%	Greater turnover and broader participation

Source : Deutsche Bank Research

The most immediate market impact may be a gradual migration from NDFs toward deliverable KRW markets. Improved access and offshore settlement infrastructure should support greater use of deliverable KRW instruments over time, although NDFs are likely to remain an important part of the ecosystem. The goal is not to eliminate NDFs, but to create more attractive deliverable alternatives, deepen liquidity and strengthen the link between offshore activity and Korean financial markets.

The implications may extend beyond the currency itself. A more internationally accessible financial system could support deeper capital markets, stronger liquidity and a broader investor base for Korean assets. Over time, this may help narrow valuation discounts that persist despite Korea's strong industrial position, improving the linkage between the country's economic performance and financial-market performance. In this sense, internationalization should be viewed not merely as an FX-market reform, but as part of a broader effort to align Korea's financial influence more closely with its growing economic and industrial importance.

Ultimately, Korea's post-war success was built on industrialization. Re-industrialization, AI investment and supply-chain diversification suggest that story is far from over. The next decade may be defined by a different question: whether Korea can convert industrial power into financial power. The success of internationalization will therefore be judged not by reserve-currency status, but by whether Korea narrows the gap between its economic footprint and its financial footprint. If industrialization was the first stage of Korea's development story, internationalization may prove to be the second.

Appendix 1

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Deutsche Bank
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The Expanding Frontier of Agentic AI: A User's View

August 6, 2026

Authors

Yi Xiong, Ph.D.
Chief Economist

The most important change in AI today may not be the arrival of a single new frontier model, but the speed at which frontier-like capability is becoming cheaper. My own use of AI agents since March suggests the shift is already visible: models that once looked like expensive high-end choices are being matched by cheaper alternatives within months. This report looks at that expanding frontier and what it means for adoption.

Since I started using AI agents in March, trying new models has become something of a hobby—and partly a necessity. I could not rely on the most advanced models to power my OpenClaw Bots and other tools, not only because access is not always guaranteed, but also because they are extremely expensive for agentic usage. I therefore spent the past few months switching among more accessible and affordable alternatives, many of them models developed by Chinese labs.

The effect of each switch was often not obvious. A model might seem more reliable or require less instruction, but the improvement was difficult to separate from chance. Over a longer period, however, the change became unmistakable. My AI agents could complete the same tasks with fewer prompts, use tools more effectively and travel further from a simple instruction before requiring intervention.

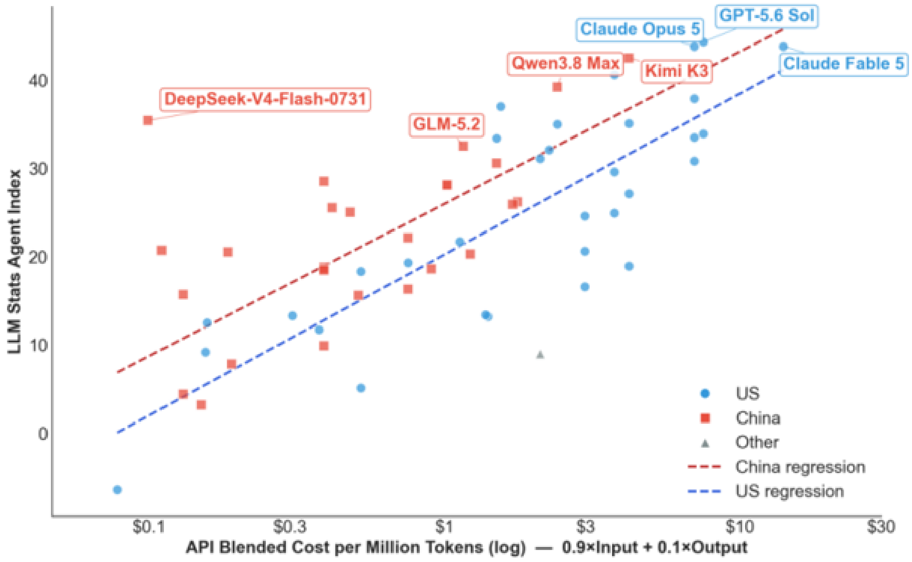
That said, two recent model releases did feel different to me. Both Kimi-K3 and DeepSeek-V4-Flash-0731 seemed notably more powerful than the previous alternatives I had been using. Was this merely a subjective experience, or had the underlying models genuinely improved? This prompted me to do some databased analytical research.

Measuring agentic intelligence against cost

The research I did was a simple plot of popular models along two dimensions: capability and cost. The horizontal axis is API cost per million tokens on a log scale, using a simple blend of 90% input tokens and 10% output tokens. The vertical axis measures model capability. There are multiple possible measures, but I chose the Agent Index from LLM Stats for two reasons: first, I care more about agentic capability than other dimensions such as coding or writing because of my use case; second, LLM Stats offers a composite measure based on benchmarks involving tool use, multi-step execution, web search, terminal operation, software engineering and other agentic tasks.

The broad pattern is intuitive: more capable models tend to be more expensive. US models are shown in blue, and Chinese models in red in the chart. A clear upward correlation can be observed across all the dots. The difference is that US models occupy much of the upper-right part of the chart, while Chinese models are more densely represented at lower price points. The regression lines capture this difference: both groups exhibit a positive relationship between cost and agentic performance, but the US cluster generally extends further into the highcost, high-capability region.

US and Chinese AI Models copared on cost and Agentic capability



Source: Deutsche Bank Research, LLM Stats

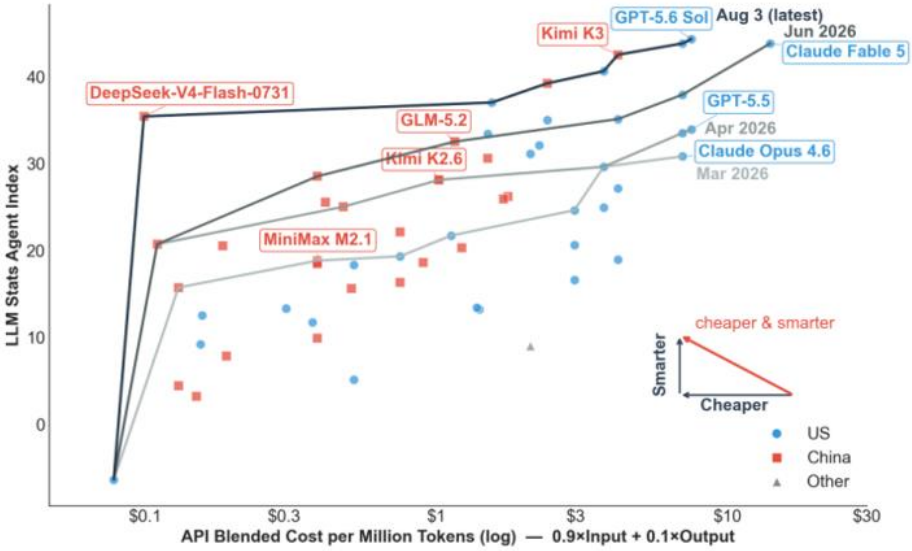
A frontier moving rapidly

The more useful concept for model selection—at least for me as an economist— is the Pareto frontier. A model is Pareto optimal if no cheaper model has a higher score. Connecting all Pareto optimal models gives the Pareto frontier: the boundary of best available model performance at each cost level. A model below that frontier may still suit a particular task, but on these two dimensions it is dominated by another choice. The chart below shows the Pareto frontier for all models in my dataset.

For example, at the top of the current frontier, GPT-5.6 Sol has the highest Agent Index in the dataset, at 44.3, with a blended cost of $7.50 per million tokens (the dot next to it is Opus 5). This is just too expensive for me. If I want to reduce cost, the next best choice according to the chart would be Kimi-K3: with only a modest loss in capability, it offers a meaningful cost saving and could be a good choice for slightly less complex tasks. But it is still too expensive for simple tasks, for which I would go even further down the line to maybe DeepSeek-V4-Flash.

But the analysis did not stop there. The chart also shows how the Pareto frontier has shifted as new models became available. By including only models released by each cutoff date — March 31, April 30, June 30 and August 3 — I was able to reconstruct earlier frontiers and compare them with the latest one.

Expanding Pareto Frontier: Agentic AI is becoming cheaper and smarter at speed



Source: Deutsche Bank Research, LLM Stats

The speed of Pareto improvement is striking. Every month, the Pareto frontier moves upward and left by a large margin. This means users can achieve better performance at the same cost, or cut spending significantly while maintaining similar performance. For example, GLM 5.2 on the June frontier has almost the same score (32.5) as Opus 4.7 (33.5) on the April frontier, while its blended cost of $1.16 is far below Opus 4.7's $7. In merely two months, a user could potentially reduce token cost by more than 80% without losing performance.

The latest frontier line makes the point even more clearly. As of today, almost all models on the latest Pareto frontier have exceeded GPT-5.5's score. In other words, **what sat on the high-cost frontier in April is now available across the cost spectrum.**

Chinese AI models are catching up

It would be easy to summarize the chart by saying that US models remain ahead… At the absolute frontier, that is still true: GPT-5.6 Sol holds the highest Agent Index in the dataset. But that reading misses the more consequential competitive dynamic.

…but Chinese models' advancement from the lower-left is also remarkable. In March, when I first started using AI agents, the Pareto frontier was dominated by US models. Kimi K2.6 moved into the middle by April; GLM-5.2 approached the performance of recent leaders by June; and Kimi K3 narrowed the gap near the top in July. Kimi K3, being the largest (2.8T parameter) and most costly openweight model to date, shows that Chinese models are no longer competing only on price; they are increasingly present near the capability frontier as well

The DeepSeek Outlier

What strikes me most is how the latest DeepSeek release altered the shape of the chart. DeepSeek-V4-Flash-0731, released last week, was a large improvement over the earlier low-cost version included in the April frontier. The wearlier model was already inexpensive, with an Agent Index of 20.7 at $0.11 per million tokens. The July release has effectively the same cost, but its Agent Index rises sharply to 35.4. It moves DeepSeek from the low-cost edge of the chart into the performance range of much more expensive frontier models.

DeepSeek demonstrates how much potential remains for AI models to improve. DeepSeek-V4-Flash is far smaller than models near the current frontier. At 284B parameters in total (13B active under a Mixture-of-Experts architecture), its agentic capabilities outperformed much larger models such as the 744B GLM-5.2 and the 1.6T DeepSeek-V4-Pro. The improvement was achieved not through increasing model size (making it more knowledgeable) but through post-training (teaching it to better handle tasks). The contrast illustrates that good agent performance does not require the largest model; architecture and post-training can move the effective frontier more than parameter count alone would suggest.

The Jevons Paradox is Working on Me

Another interesting fact: the steep decline in AI model costs did not bring me any savings. The opposite is happening—my AI bills have surged. My total spending on AI was effectively zero in February before I started using AI agents. Then I started using OpenClaw and made my first token subscription plan at $20/month. A few months later, my recurring spending on AI is now reaching $100-200/month. My costs increased because I now use AI agents to do many more things and must subscribe to new plans as tokens run out under existing ones. The increase also includes cloud storage and an $800 Mac Mini to host my “little lobsters” (OpenClaw Bots).

This is a familiar economic mechanism: the Jevons paradox. Improvements in efficiency did not reduce consumption; rather, they encouraged wider use and led to an increase in total usage and total spending. As model capability improved while token cost dropped, I expanded the use of my AI agents to more daily tasks—tracking headline news and specific topics I am interested in, summarizing market movements, researching academic papers, managing expenses, making travel plans and testing ad-hoc ideas. Why use a chatbot if you have dedicated AI agents that know your background and remember the conversation history?

The question is therefore not simply which model will hold the highest score next month. It is what happens when millions of users and businesses can afford to keep capable agents running for longer, on more tasks, with less concern about every token they consume. If the cost of agentic intelligence continues to fall while its capability rises, will efficiency save tokens—or will Jevons’ paradox help trigger an explosion in real-world AI use?

My verdict: the expansion of the AI Pareto frontier is benefiting users and the broader AI ecosystem. As US and Chinese labs compete, premium frontier models push the performance ceiling higher, while efficient lower-cost models make agentic workflows economical at scale. Together, they allow users to match each model to the value and difficulty of the task, rather than accept a single price-performance trade-off. Competition is accelerating the race, while the global circulation of research ideas, open weights and practical techniques is helping improvements spread quickly across the field. **The analysis made me smore confident that the future of agentic AI is arriving faster than many expected.**

Appendix 1

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Deutsche Bank
Research Institute

Situational awareness - do we have it?

August 3, 2026

Authors

Luke Templeman
Research Analyst

Galina Pozdnyakova
Research Analyst

Executive summary

There are two events in the last few weeks that look like non-events. The Situational Awareness hedge fund unwound a $16bn book of leveraged AI trades with barely a ripple beyond a few days of volatility. Meanwhile, the Kospi fell more than 20% in 48 hours and then rallied 25% off the low — finishing the week almost flat. Taken individually, both look like markets doing their job: absorbing shocks and moving on.

We think that read is wrong, or at least incomplete. Both events were symptoms of the same underlying condition: leverage in ETFs, margin accounts and hedge fund books that was encouraged by more than a decade of low and stable rates. Indeed, reports that over 3% of Korea's adult population received a margin call over the last two weeks is deeply concerning if true.

This piece argues that 'small vol' events are going to affect markets with an increased frequency. That comes as the backdrop of normalised interest rates and geopolitical unrest has been augmented by a Fed that has moved away from the forward guidance that reduced some volatility in the past.

We also worry that the risks in household debt (and investment accounts on margin) is being ignored as the focus stays on the risks in growing sovereign debt. If this is right, 'small vol' episodes are likely to become more frequent and potentially have larger consequences.

In the end, we worry that investors, corporates and policymakers are focused on avoiding the next big financial crisis and, in doing so, may be underestimating the risk in small events that can cascade.

A tale of two 'small vol' events

Markets have spent the last few years braced for another 2008-style event. That vigilance has a blind spot. It trains attention on the big, obvious risks and away from the smaller, everyday ones that can cascade in the right circumstances.

The implosion of the Situational Awareness hedge fund is the first case in point. The fund sold its $16bn public equity book after highly leveraged trades on AI companies went wrong. Markets absorbed the unwind with only a brief bout of volatility — no contagion, few headlines beyond the financial press.

The second was in Korea. At one point last week, the Kospi fell over 20% in barely 48 hours, before staging a 25% rebound from the low. The headline index ended the week close to flat and broader markets have shrugged it off. But a flat index conceals an extraordinary amount of forced selling and buying beneath the surface, much driven by margin calls on leveraged retail positions.

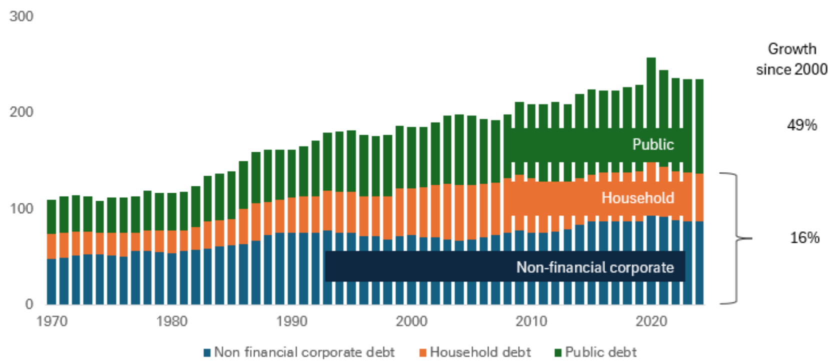
Why ‘flat on the week’ is the wrong takeaway

An index that round-trips 45 percentage points in days is not the same thing, economically, as one that simply sits still. Leveraged and margined positions are typically closed out mechanically and irreversibly on the way down, regardless of what happens on the way back up. The P&L may net out at the index level; the wealth destruction for individual, often first-time, leveraged investors does not.

We believe the potential for household leverage to cause problems is underestimated because the aggregate figures look okay. When we show the following chart, clients gravitate to the rise in sovereign debt, not the plateau in household leverage.

Figure 1: Household debt has lost the spotlight to public debt since 2000

Global debt as a % of global GDP

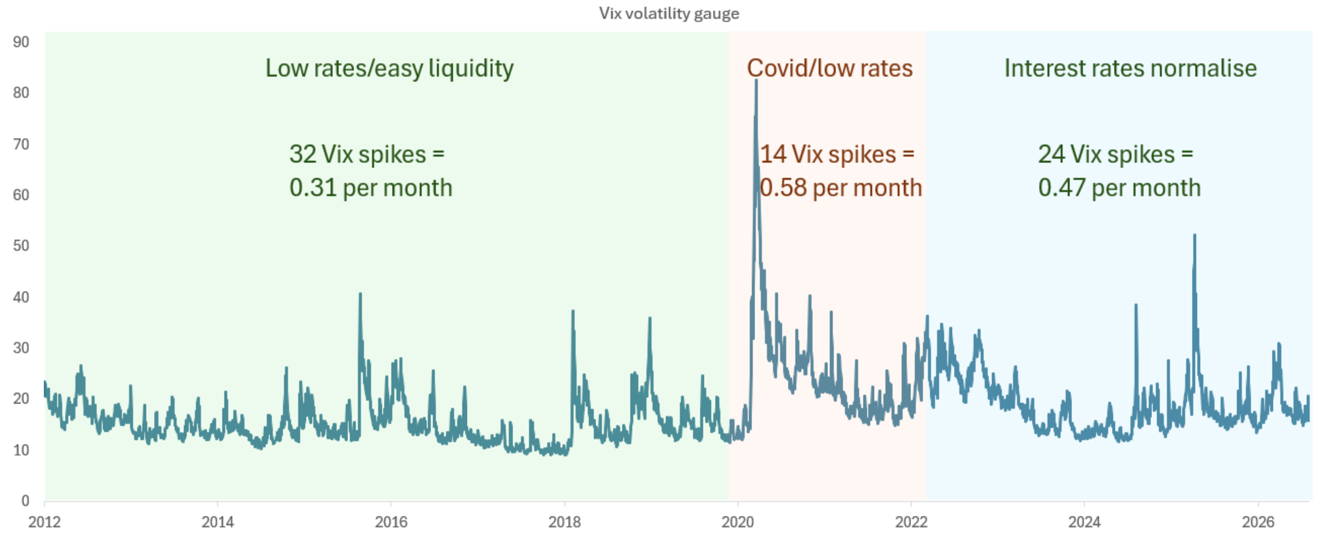


Source: IMF, Deutsche Bank Research

Small vol events are becoming more frequent

The following chart is created using our own judgement. But it shows that, by our reckoning, the number of spikes in the Vix since 2022 has been materially higher over the last few years than during last decade. The last four years are an important comparison period as this is when interest rates began to normalise.

Figure 2: A greater frequency of 'small vol' events appears to be the norm, at least for now



Source: Factset, Deutsche Bank Research

Is the Fed trying to bring volatility back on purpose?

One explanation for why policymakers may be tolerating episodes like these comes from Rob Armstrong (a former colleague) here. He advanced a theory about Fed chair Kevin Warsh that essentially says, his withdrawal of forward guidance is a deliberate attempt to reintroduce 'uncertainty premia' into markets that have used lower volatility to take on more leverage than they otherwise would. The idea is that "Too little everyday volatility ultimately leads to financial crises."

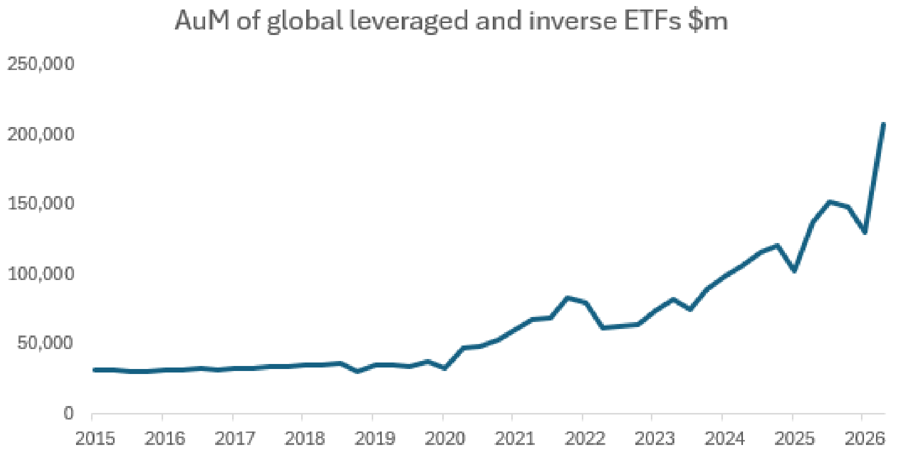
The mechanism, on this reading, is that forward guidance suppresses everyday volatility — and stable prices and calm markets encourage more borrowing, both by governments and by market participants, than would occur under normal price discovery. Warsh cannot say this outright without spooking markets, hence the vaguer 'referee' rhetoric — but the policy actions, in Armstrong's view, point the same way.

If policymakers genuinely believe more day-to-day volatility is healthy, investors should expect more 'small vol' episodes like Situational Awareness and the Kospi swing and should stop treating each one as isolated, ring-fenced events.

Leveraged ETFs and margin debt: Should we be afraid?

Leveraged and inverse ETFs, and the margin debt that often sits alongside them, are growing quickly across many major markets. They are a key mechanism that turns an ordinary correction into a cascade because they force mechanical, non-discretionary selling as prices move.

Figure 3: The semiconductor and broader tech rally has fuelled the popularity of leveraged ETFs



Source: Bloomberg Finance L.P., Deutsche Bank Research

In Korea specifically, the launch of single-stock leveraged ETFs tracking Samsung Electronics and SK Hynix in April turned two already-dominant, trillion-dollar-plus stocks into vehicles for retail leverage almost overnight. Retail investors accounted for a large proportion of holders. When the underlying stocks fell, those products amplified the move.

The risk is not confined to Korea. The same products are growing in the US and Europe. A market that is many times larger, and far more directly connected to consumer spending and credit availability, having its own 'Kospi moment' is a materially bigger problem for systemic risk than a niche hedge fund unwinding.

A footnote: why the name 'Situational Awareness' went viral

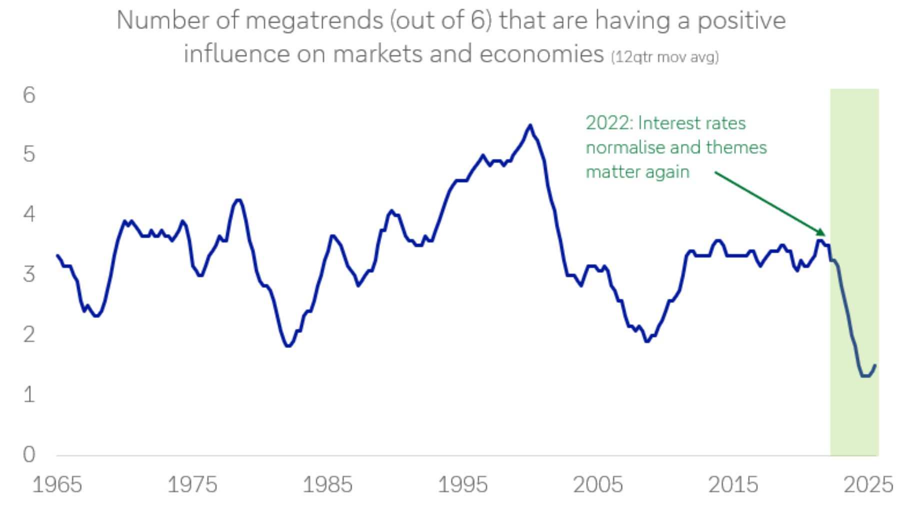
The fund's name borrowed from Leopold Aschenbrenner's 165-page 2024 essay, which argued AGI was plausible by 2027 and sketched a path from massive compute build-outs to an 'intelligence explosion' and eventual government-led consolidation of frontier AI ('The Project').

Part of a developing medium-term trend

Our megatrend framework tracks six long-run forces (technology, sovereign deficits, geopolitics & globalisation, domestic politics & social discontent, demography, and energy) and combines them into a single indicator of how supportive the backdrop is for markets and economies.

Overall, the number of megatrends working in favour of markets and economies has fallen to a level last seen only twice before — during the 1970s oil crises and around the 2008 financial crisis. Both were periods marked by exactly this kind of 'small vol' pattern: idiosyncratic-looking events that, in hindsight, were symptoms of a more fragile underlying system.

Figure 4: Prior periods of megatrends being this deeply negative have resulted in the greater frequency of volatility events



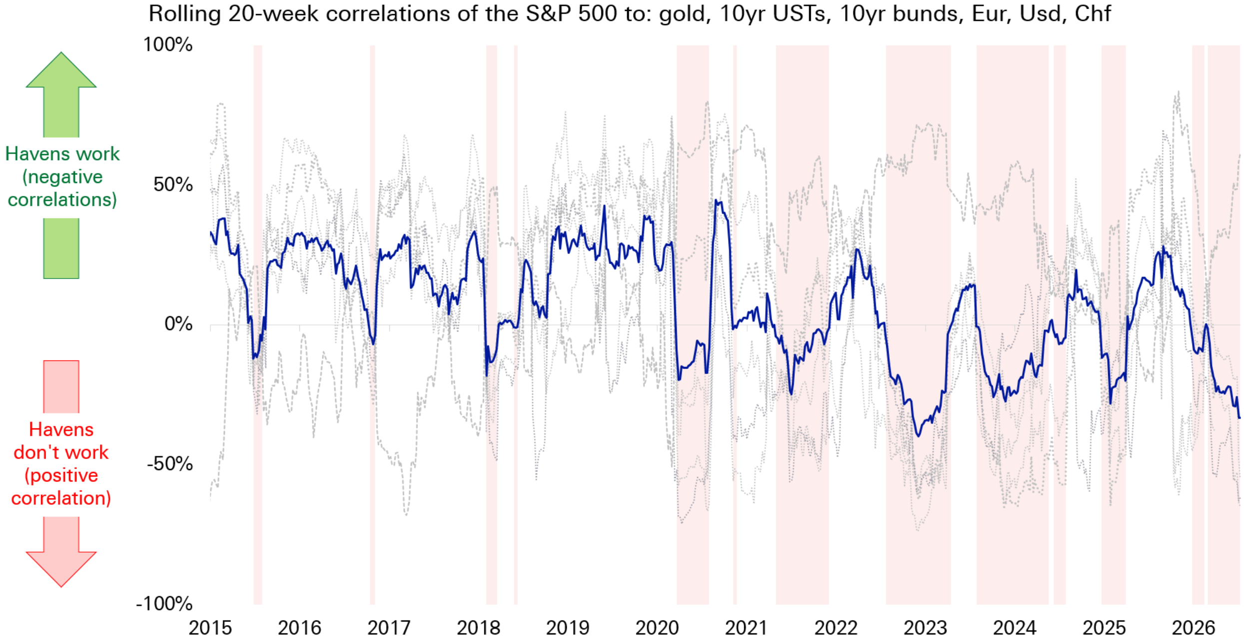
Source: Deutsche Bank Research

The single most important negative megatrend is sovereign deficits. Developed-market governments are refinancing and issuing new debt at rates well above those of the last decade. At the same time that our model flags unpredictable sovereign debt sell-offs, particularly those triggered by exogenous shocks and investor emotion, as a growing risk.

Havens are not there to catch you

Ordinarily, investors would look to traditional havens (gold, the dollar, the Swiss franc, the yen, US treasuries and bunds) to cushion volatility. However, our haven indicator shows that, since 2020, this combination of assets has failed to hedge risk (as proxied by the S&P 500) particularly during more shock events, including covid, the 2022 rate hikes, the 2025 tariff shock and the 2026 Iran war.

Figure 5: Traditional haven assets have not worked in the 2020s, particularly during shock events and periods of volatility



Source: Bloomberg Finance L.P., Deutsche Bank Research

If havens keep failing when they are needed most, the cushioning effect investors implicitly rely on to keep 'small vol' small is weaker than it used to be. That raises the odds that future 'small vol' episodes may cascade into something more.

What if this happened in the US or Europe?

It is tempting to treat the Kospi swing as a Korea-specific story — a function of retail culture, single-stock products and two dominant, correlated semiconductor names. But if a similar episode plays out in US or European, the results could be systemic.

The transmission mechanism would not be limited to portfolio losses. It would run through consumer spending, as households with margin-called leveraged positions cut back; through credit, as brokers and lenders reassess collateral and risk; and through confidence, as retail participation reverses. A leveraged retail shock in the US would be a materially larger systemic risk than a single hedge fund unwind.

Another example

Silicon Valley Bank is another example. On paper, SVB's failure should have been a contained, idiosyncratic event — a mid-sized bank with a concentrated depositor base and a duration mismatch. In practice, it triggered enough of a confidence panic that policymakers felt compelled to effectively guarantee all deposits across the banking system, a far larger intervention than the initial problem seemed to warrant.

The lesson is not that every small event becomes systemic but the ones that do are rarely obvious in advance. And when there is an increased frequency of these events in a market with more leverage, less reliable havens, and policymakers newly tolerant of everyday volatility, there is an increased risk that 'small vol' events may catalyse larger problems.

Conclusion: are we situationally aware?

The market's instinct is to file Situational Awareness and the Kospi swing under 'noise' because neither, so far, has cascaded into a systemic event. That instinct may be dangerous, particularly if policymakers are willing to tolerate more everyday volatility.

Given our megatrend model indicates that the supportive backdrop for markets is near a multi-decade low, 'small vol' episodes can potentially be used as an indicator of how stress in the financial system is quietly building up.

There are plenty of other parts of the market where 'small vol' events may catalyse problems in areas that were built up with leverage during the periods of easy money. Just two examples are:

- Private equity: a large amount of TMT portfolio companies bought during the era of near-zero rates remain on PE firms' books, held longer than usual as exit markets stayed shut. A reckoning may still be ahead.
- Corporates: many listed companies still carry underperforming divisions bought during debt-fuelled M&A booms last decade. As a result, they have lower asset turnover metrics, a legacy of a decade in which cheap financing incentivised growth in revenue over earnings.

Appendix: Why Situational Awareness went viral

The original essay by Leopold Aschenbrenner in 2024 was 165 pages long. Below is a bullet point summary (generated by AI).

The Path to AGI by 2027: The document argues that Artificial General Intelligence (AGI) is strikingly plausible by 2027. This prediction is based on extrapolating "Order of Magnitude" (OOM) trendlines in three areas: physical compute scaling, algorithmic efficiencies, and "unhobbling" gains (transforming models from chatbots into active agents).

Massive Industrial Mobilisation: AI is shifting from a software boom to a massive industrial process. Leading AI labs will build training clusters costing $100s of billions by 2028 and over $1 trillion by 2030. By the end of the decade, these clusters could require power equivalent to 20% of total U.S. electricity production.

The Intelligence Explosion: Once AGI is achieved, it will likely be used to automate AI research itself. Because digital researchers can run in millions of copies and at much higher speeds than humans, a decade of algorithmic progress could be compressed into a single year, leading rapidly from AGI to superintelligence.

The Security Crisis: Current security at leading AI labs is described as "Swiss cheese," leaving key algorithmic secrets and model weights vulnerable to stateactor theft, particularly by the CCP. The author warns that leaking these secrets in the next 12-24 months would be a catastrophic national security failure.

Superalignment Challenges: Reliably controlling AI systems that are vastly smarter than humans is an unsolved technical problem. Current alignment techniques (like Reinforcement Learning from Human Feedback) will break down because humans cannot supervise behaviour they do not understand. Solving this will require using AI to help automate alignment research.

"The Project" (U.S. Government Intervention): Given the decisive military and economic advantage of superintelligence, private startups cannot handle it alone. By 2027 or 2028, the U.S. government is expected to initiate a nationalized AGI effort—referred to as "The Project"—to consolidate resources, secure model weights, and ensure the free world prevails over authoritarian powers.

Geopolitical Stakes: Superintelligence will provide a military edge comparable to nuclear weapons. A lead of even a few months could be decisive, making a healthy lead for democratic allies essential for safety and global stability.

Appendix 1

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Research

This Month in Geopolitics: August 2026

Key events, themes and trends to watch in the month ahead

August 2026

Dr. Helen Belopolsky | Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in **August 2026**

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.

United Kingdom
- **All August:** New Prime Minister Andy Burnham is likely to set the tone for his premiership through early statements on cost-of-living measures and his forthcoming ten-year "new plan for Britain".

Iceland
- **29 August:** Iceland will hold a referendum on resuming EU accession talks in light of growing global geopolitical and trade frictions.

Kazakhstan
- **23 August:** Kazakhstan will hold **legislative elections** for the newly created Kurultai, the first election to this new unicameral legislature. This is the first change to the legislative structure in three decades.

US/Mexico/Canada:
- **All August:** US-Mexico negotiations will focus on resolving outstanding issues while negotiations with Canada will be complicated as a result of a new US 50% tariffs on Canada and the Canada-US 30-day countdown for a new cross-border economic/security agreement.

China:
- **Early August:** China's leadership will meet for its summer retreat in Beidaihe. The event provides an opportunity for Party leaders to exchange views on major policy issues.

United States:
- **All August:** The Trump administration will likely keep building its replacement "permanent tariff framework" with a number of Section 301 investigations signaling further tariff actions in the coming weeks.
- **All August:** China-US talks ahead of the next Xi-Trump summit will be a key watchpoint for potential deliverables on trade, investment and AI dialogue.
- **31 August to 1 September:** The G20 Meeting of Finance Ministers and Central Bank Governors may be contentious.

Colombia:
- **7 August:** Abelardo de la Espriella will be inaugurated as Colombia's next president. His administration has vowed significant changes to Colombia's domestic and foreign policies.

US-Iran:
- While continued US and Iranian strikes and bellicose rhetoric threaten an escalatory spiral, mediators and regional actors will continue shuttle diplomacy towards a negotiated offramp.

Source: Deutsche Bank

In the spotlight for August 2026: US-Iran peace talks and US trade policy

US-Iran war (all of August)

Expect more volatility – but also efforts to resume dialogue – during August. Tensions escalated significantly in July, with renewed military strikes and the US reimposing sanctions on Iranian oil sales. Both sides appear unwilling to concede for fear of losing ground. While President Trump has suggested that the ceasefire is over, **we retain our assessment that both parties are incentivized to find a near-term resolution** as kinetic attacks will not drive a resolution. That said, we should prepare for more volatility in the coming months as Iran is unwilling to cede control of the Strait of Hormuz and neither side is prepared to lose face. The Yemen-based **Houthi blockade of Saudi Arabia**, announced on 20 July, further heightens risks by threatening crude oil flows via Saudi Arabia's Yanbu on the Red Sea (see map), a terminal which handles about 4 million barrels of crude oil and condensate exports per day.

The persistent security risks will likely continue to drive the Gulf states to reduce their vulnerability to the Strait of Hormuz. Shipping levels through the Strait will periodically be impacted by Iranian threats to navigation and political considerations. Even once dialogue resumes and fighting subsides, **traffic will likely take time to recover** given the challenges of de-mining, a lack of trust between the US and Iran, and Iranian threats. Infrastructure repair to ports, terminals and energy infrastructure, as well as the provision of security guarantees are other factors to watch. Full operational normalization, involving infrastructure repairs, is not expected until 2027 and beyond.

US trade policy (all of August)

When the US 10% global Section 122 tariffs expired **on 24 July**, the Trump administration moved quickly to fill the gap with Section 301 tariffs of 10-12.5% on more than 60 countries. In August, we should expect the Trump administration to **keep building its replacement "permanent tariff framework"** using Section 301 and 232 authorities. The USTR's Section 301 investigations into "structural excess capacity" in 16 manufacturing-heavy economies remains ongoing and could yield new tariffs in the coming weeks. The Trump administration is also pursuing several Section 232 national security investigations with scheduled tariff effective dates of 31 July and 29 Sept. This should keep pressure on countries to strike bilateral deals favorable to the US, spanning trade, investment, industrial and technology concessions.

To watch will be movement on unresolved country-specific issues, e.g., India's tariff talks, the EU's proposed 30% tariff, in addition to continued legal uncertainty as court appeals over the tariffs' legal basis play out. This will set a complex backdrop to the G20 finance ministers' meeting where trade protectionism is already shaping up to be a point of friction.

Key regional ports in the Persian Gulf

Source: Deutsche Bank. For more on the impact of the Iran conflict, see Twenty Miles That Shook the World - Deutsche Bank Research Institute

Key events, issues and themes to watch in **August 2026**

Listed in chronological order

US-Mexico-Canada trade talks (all of August)

The renewal talks are expected to be challenging and potentially protracted. 1 July marked the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process whereby the three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode. The US has declined to confirm the renewal of USMCA "in its current form".

In August, we expect further US-Mexico-Canada negotiations focused on resolving outstanding issues (e.g., automotive rules of origin, steel/aluminum, economic security, and issues around agriculture, labour and environment). At the G7 summit, Canada and the US agreed to a 30-day countdown for a new cross-border economic/security agreement, which was extended to 1 August.

In any case, Mexico and Canada are both subject to a 10% Section 301 tariff that took effect on 24 July. In addition, the US 50% tariffs on a range of Canadian goods announced on 20 July, and additional Section 232 investigations (lumber, copper, semiconductors, pharmaceuticals, trucks, critical minerals, etc.) could complicate talks further as Mexico and Canada are top exporters in several of these sectors. The Trump administration is also using the review to push non-trade demands on migration, drug trafficking and defence cooperation.

First weeks of new UK Prime Minister (all of August)

Andy Burnham became the UK's 7th Prime Minister since 2016 on 20 July. He enters office facing a sluggish economy, cost-of-living pressures, and a massive welfare bill. He has pledged this moment as a "circuit-breaker" for Britain and has quickly set out to focus on concrete cost of living measures. In August, we can expect Mr. Burnham to be busy with the early scaffolding for his "No. 10 North" project focused on devolution of power. Mr. Burnham has signaled his priorities will be cost-of-living issues, devolution, public control of essentials, social commitments and longer-term structural reform. UK parliament is typically in summer recess through August, which limits legislative action but gives Mr. Burnham room to set the tone for his premiership with an intention to move quickly on key commitments in the autumn.

Prime Minister Andy Burnham was sworn in on 20 July. Source: House of Commons

Key events, issues and themes to watch in **August 2026**

Listed in chronological order

China-US talks ahead of next Xi-Trump summit (all of August)

Bilateral dialogue to tee up deliverables for the 24 September summit will be a key watchpoint during August. Recent developments – including US allegations of election interference and warnings over Chinese AI models – reinforce our view that relations remain structurally competitive despite the recent stabilisation in ties. Investors should anticipate more, not fewer, bilateral economic restrictions over the medium term.

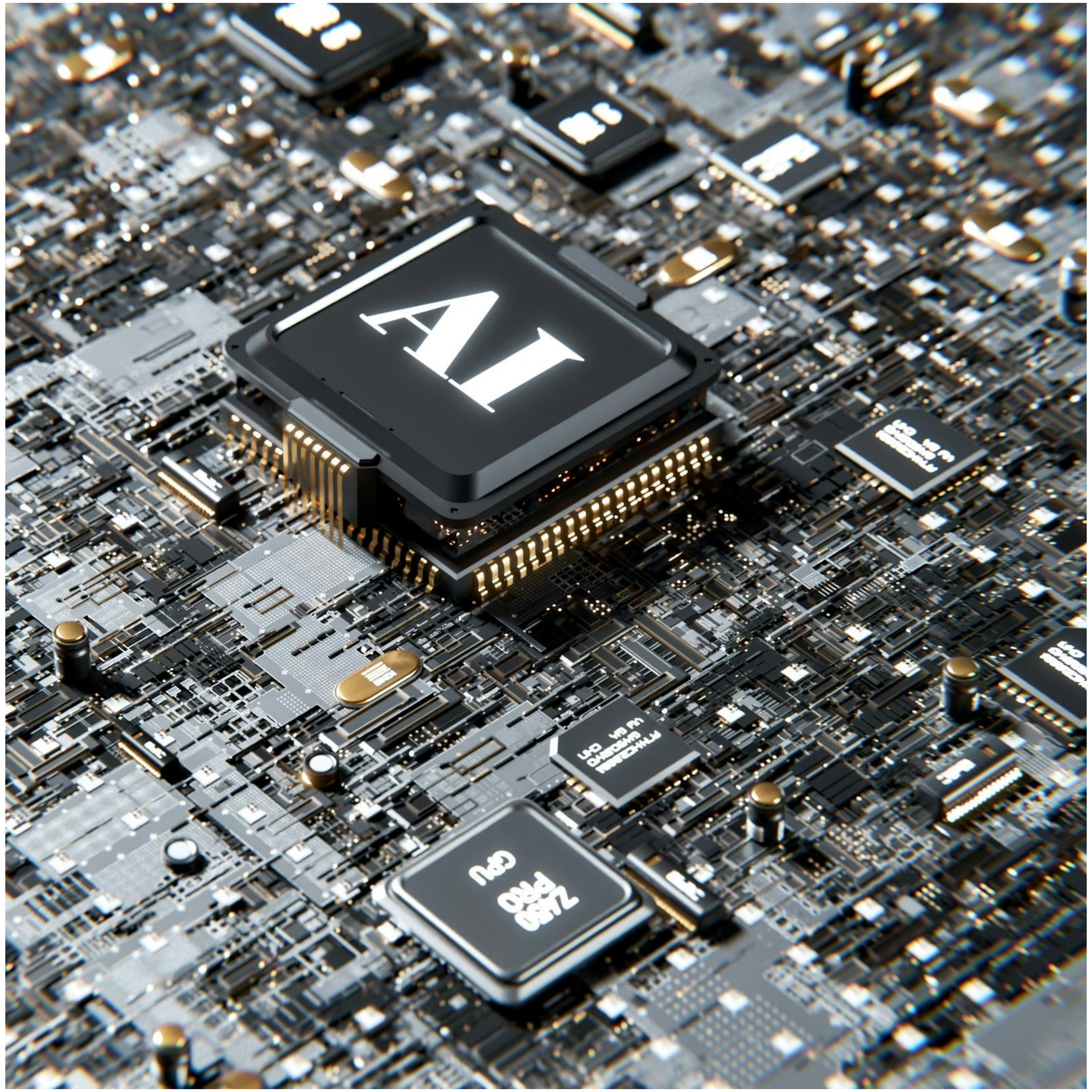
Nevertheless, there is currently no evidence to suggest these dynamics are likely to delay the next Xi-Trump summit, which will likely go ahead as planned on 24 September. During August, investors should watch for any sign of progress on talks on the Board of Trade (intended to boost non-sensitive trade), Board of Investment (to facilitate investment in non-sensitive areas) and an extension of the Busan agreement – which delayed a range of new economic restrictions until November 2026.

Both sides could also strengthen commitments around AI dialogue, with first talks reportedly expected to happen ahead of the summit. Dialogue on AI was a key deliverable from the May Xi-Trump summit, and both sides could seek to address persistent tensions around the technology, although a major breakthrough appears unlikely.

China's leaders gather for summer retreat in Beidaihe (early August)

The retreat provides an opportunity for Party leaders to exchange views on major policy issues. This year's meeting could potentially cover topics ranging from the conflict in Iran, relations with the EU and US, the implementation of the 15^{th} Five-Year Plan (2026-2031) and early deliberations around the next Party Congress in late 2027.

The retreat primarily serves as an opportunity for internal alignment and consensus building on key domestic and external policy issues. There are no official readouts, but subsequent state media framing, leadership signals, and Party meetings in the autumn could offer clues on the direction of policy debate.



Source: Igor Omilaev on Unsplash

Key events, issues and themes to watch in **August 2026**

Listed in chronological order

Abelardo de la Espriella inaugurated as Colombia's next president (7 August)

The president's early agenda is expected to include significant policy shifts, including a planned urban security coordination mechanism for major cities; a more pro-US and pro-Israel foreign policy stance; and a review of Colombia's diplomatic posture toward Cuba, Nicaragua, Venezuela and multilateral bodies. Key watchpoints over the next 2-3 months will include cabinet formation and congressional coalition-building. The incoming administration has also vowed to disrupt narcotics production and trafficking, and counter violence by armed groups operating in parts of the country – including in a bid to attract more foreign investment.

Kazakhstan legislative elections (23 August)

Kazakhstan will elect all 145 deputies to the newly-created Kurultai. This will be the first election to the new unicameral legislature established after March 2026 constitutional reforms replaced the old bicameral parliament. These elections are unlikely to materially impact governance in the country with President Tokayev remaining firmly in control of the political system. Kazakhstan's constitutional court ruled that president Tokayev may seek another term under the new 2026 constitution, effectively resetting his term count despite having been elected in 2022 to what was billed as a single, non-renewable seven-year term ending 2029. While the election itself is unlikely to produce policy surprises, it will be an indicator of the government's ability to manage the political transition under the new constitutional framework and maintain domestic stability.

Iceland referendum on resuming EU accession talks (29 August)

Iceland will hold a referendum asking voters whether the country should resume EU accession negotiations. Iceland had applied to join the EU in 2009, but suspended accession talks in 2013 after a change of government and later withdrew the application.

The current coalition government pledged to finally let voters decide, and accelerated the timeline partly due to shifting geopolitical dynamics, including US tariff hikes and Trump administration statements on acquiring Greenland. Polling is tight with surveys showing 52% in favor with 28% opposed. A vote to resume negotiations would not commit Iceland to EU membership but would mark a significant shift in the country's strategic orientation.

Abelardo de la Espriella will be inaugurated as Colombia's next president on 7 August. Source: Wikimedia Commons

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Key events, issues and themes to watch in **August 2026**

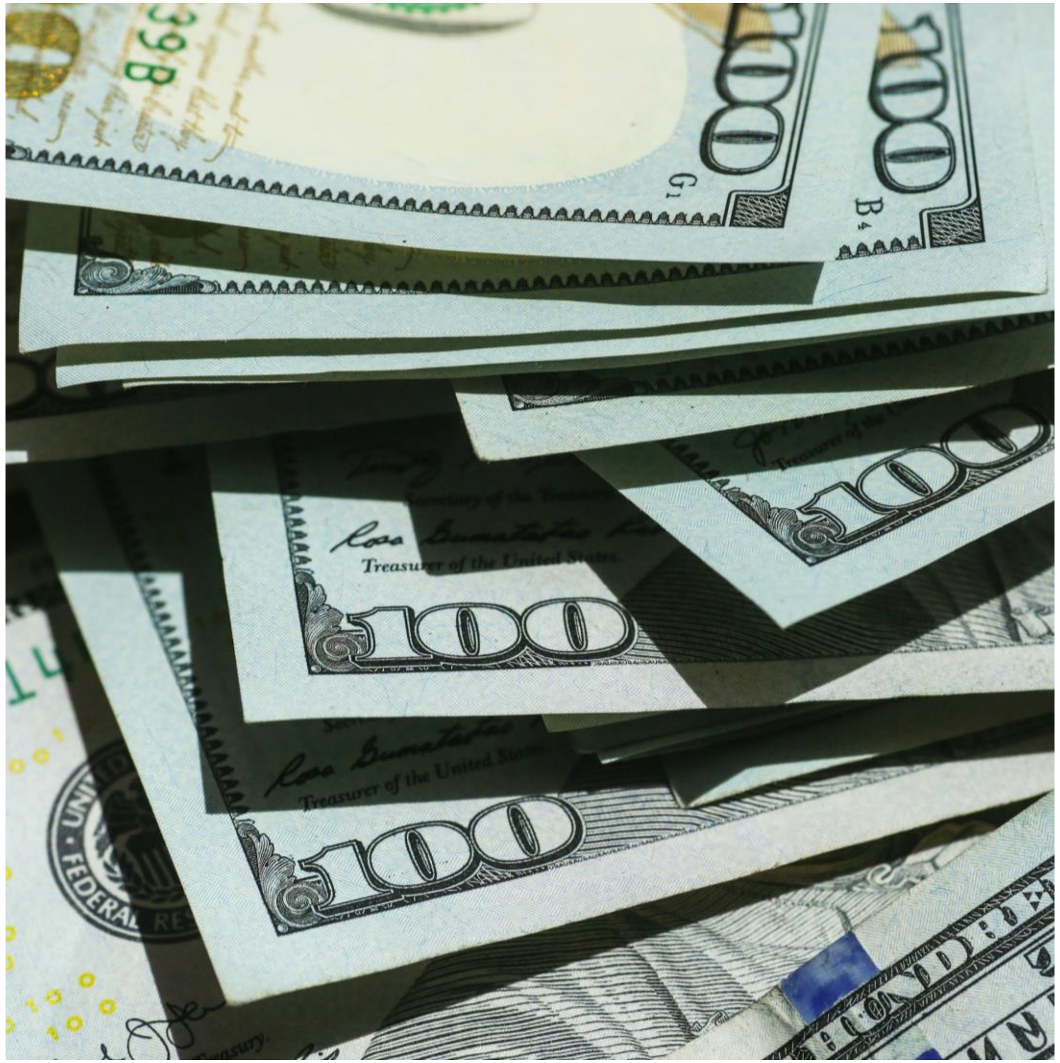
Listed in chronological order

G20 Meeting of Finance Ministers and Central Bank Governors (31 Aug to 1 Sep)

This will be a key checkpoint in the US' G20 presidency, where Washington sets the agenda for global economic policy coordination among the world's largest economies. The US has identified as its priorities: modernizing financial regulation, strengthening understanding of excessive global imbalances, enhancing debt transparency and facilitating debt restructuring, endorsing digital assets and improving cross-border payments. Treasury Secretary Scott Bessent has sought a "streamlined, results-oriented" agenda.

The first G20 finance ministers meeting under the US presidency in April failed to produce a joint communiqué or chair statement and no joint press conference was held at the end. The current crisis in the Middle East amongst other geopolitical developments may crowd out purely technical finance discussions.

Washington's tighter focus may clash with other members who still want to discuss debt relief for developing countries, climate finance and inequality. With a US leadership openly skeptical of the G20's broader mission, to watch will be how much substantive coordination is feasible.



Source: Giorgio Trovato on Unsplash

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Global events matter, but don't, but do

July 30, 2026

Executive summary

There is a common assumption that markets look through global events such as war and geopolitics. But the effects of these and other events extend far more broadly into investor portfolios than just the headline indices. Our latest data indicates just how much most people oversimplify the reaction of investors and markets to global events.

Indeed, we find that investors do not look through global events. On the contrary, they are highly polarised about them. On our analysis, the vast majority of investors think global events will either be a “risk” or an “opportunity” for markets. That is very different from almost all other broad themes where a higher proportion of people simply think there is “no effect”.

Our dbDataInsights team asked 500 people from the US and UK if and why they think global events affect their investment portfolios and what they are doing about it. We then used AI to analyse all this unstructured data. In this piece, we compare the results to specific datapoints that indicate investors are actively reshaping their portfolios to either capture opportunities or hedge the risks that they see in global events.

In the end, we see that positive and negative views on global events can coexist. The negative respondents focus on the immediate effects: oil prices rise, inflation rises, markets fall. The positive respondents focus on longer-term effects: markets adapt, companies innovate, energy systems diversify, and portfolios positioned in the right sectors benefit. Both views may be internally coherent; they simply operate on different time horizons.

Authors

Luke Templeman
Research Analyst

Galina Pozdnyakova
Research Analyst

Why global events are making a comeback

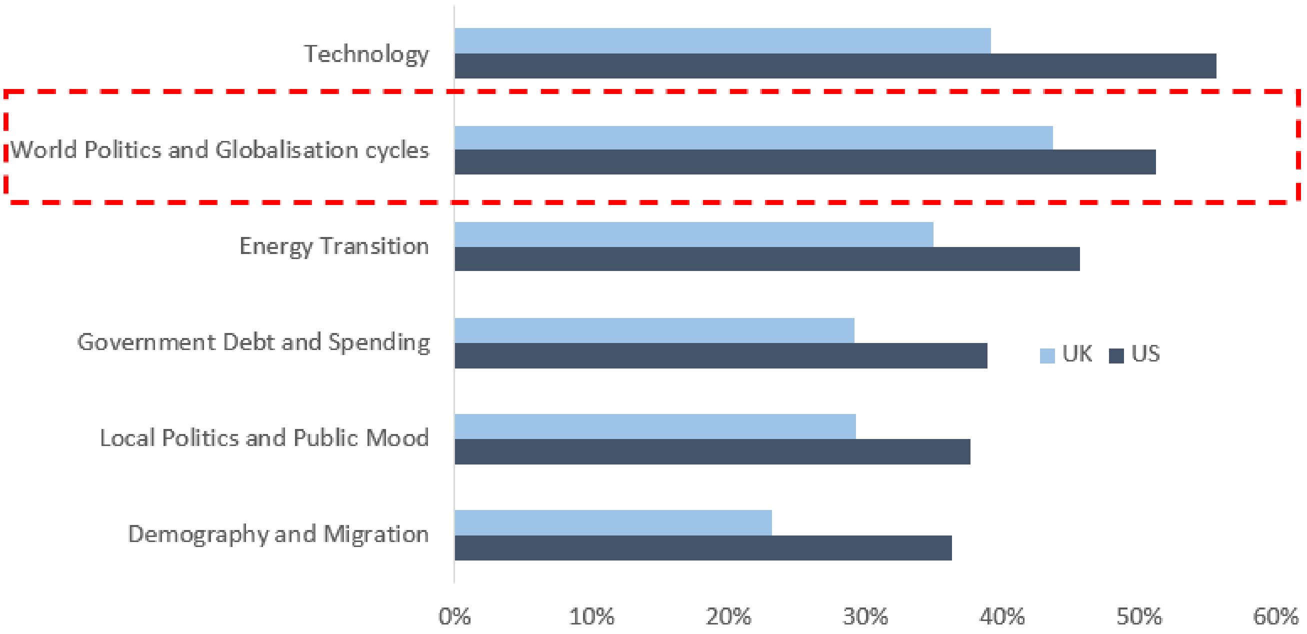
Global events are more than just war and geopolitics. They extend to incidents that affect energy prices, inflation, fiscal sustainability, and technology. Indeed, these forces are some of the strongest that affect global economies and we discuss them in detail in our piece on global megatrends.

Our thesis is that, for much of the post-financial-crisis period, low interest rates drowned out the other effects of global events. Low rates raised valuations, pushed investors into risk assets, and made it easy to explain market moves through central bank policy. Now that rates have normalised, global events are more visible as market drivers again.

Look through or not, investors are changing their portfolios

Our data indicates that investors are reshaping their portfolios as a result of global events. These investors view them not as merely one-off shocks but things that are becoming persistent forces.

Figure 1: Over the last three years, have the following thematic megatrends resulted in you making changes to your investment portfolio?



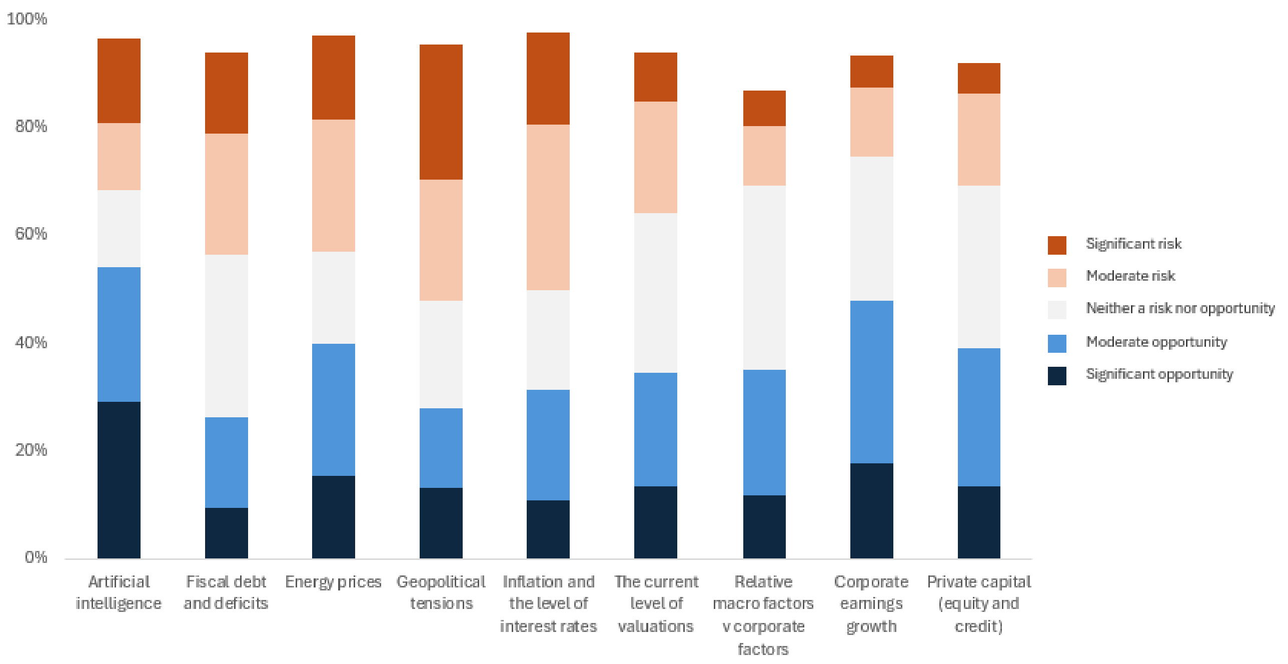
Source: Deutsche Bank dbDataInsights

The investor base is therefore not split between people who “understand” global events and people who do not. Rather, it is split between investors who emphasise different time horizons and different asset exposures. A broad equity investor may fear a Middle East shock because it threatens inflation and sentiment. An energy, gold, or defence investor may welcome the same shock because their holdings could benefit. A long-term diversified investor may view the event as noise.

Investors are highly polarised about global events

Retail investors are not passive observers as many may believe. The following table shows that our surveyed investors have strong beliefs both ways about global events and their impact on portfolios. Most importantly, when we look at geopolitics, energy prices, and inflation, the proportion of people who think there is no effect is very low compared with the other themes.

Figure 2: For each investment theme, is it a risk or an opportunity for investors in 2026? (US)



Source: Deutsche Bank dbDataInsights

Aside from the shock effect, it is understandable that the same global event can produce very different portfolio expectations. A war in the Middle East can be viewed as a risk because it raises oil prices, hurts consumers, and depresses equities. But it can also be viewed as an opportunity if an investor owns energy stocks, gold, defence shares, or believes that markets will rebound after the shock. Indeed, in our survey, several positive-impact respondents explicitly mentioned these assets and actions.

Energy is a good example. Short-term energy disruption is painful for consumers and companies. Yet, energy disruption can have positive medium-term effects if it pushes economies toward greater energy security, diversification of supply, and lower energy intensity. Indeed, since 1990 the amount of GDP generated per kilogram of oil equivalent has quadrupled in terms of purchasing power parity. In other words, economies have become much more energy efficient over time which helps explain why the current closure of the Strait of Hormuz has not yet had the disastrous effects of the oil shortages during the 1970s.

This is consistent with our megatrend framework which shows geopolitics and globalisation do not have simple linear impacts on economies. Trade in goods can be reorganised through supply chains, and trade in ideas can continue to accelerate through connectivity. In other words, geopolitical deterioration is not always a pure negative for markets. It changes the winners and losers or increases dispersion.

The most important topics

The following table shows the top 3 themes that were observed in our survey of why people think global events are either positive, negative, or neutral for their portfolios.

Figure 3: Top 3 types of global events that matter to people

Global events that make people more positive, negative, or neutral about their portfolio			
US	Negative	War and geopolitical conflict, especially Iran, Middle East, Ukraine/Russia and escalation risk	28%
US	Negative	Inflation and cost-of-living pressure reducing returns and company earnings	14%
US	Negative	Oil, gas and energy price shocks feeding through to markets	13%
UK	Negative	War and international conflict, especially Iran, Middle East, Ukraine/Russia and wider unrest	26%
UK	Negative	Inflation, rising prices and higher living or business costs	20%
UK	Negative	Energy, oil and fuel price shocks, including Middle East supply disruption	17%
US	No impact	Diversification across assets or holdings expected to absorb shocks	25%
US	No impact	Long-term investing, with short-term events expected to smooth out over time	18%
US	No impact	Low-risk, fixed income, cash-like or insulated investments	18%
UK	No impact	Long-term investing, with market ups and downs expected to smooth out	25%
UK	No impact	Uncertainty, lack of knowledge or unclear rationale	22%
UK	No impact	Diversification or resilient portfolio construction	18%
US	Positive	Global events creating growth opportunities, new markets or higher returns	20%
US	Positive	Technology, AI, digital transformation, healthcare, renewable energy or innovation	16%
US	Positive	Economic recovery, stabilisation and improving market confidence	15%
UK	Positive	Economic recovery, stronger markets and improving investor confidence	20%
UK	Positive	General optimism or confidence that things will improve	16%
UK	Positive	Global events creating investment opportunities, new markets or growth sectors	15%

Source: Deutsche Bank Research

Energy prices are the everyday face of geopolitics

Energy prices are the mechanism through which many investors experience global events most directly. In the written responses, distant conflicts become financially relevant because they raise oil, gas, petrol, utilities, shipping costs, food prices, and inflation. This is especially visible in the UK negative responses, where energy costs and inflation appear repeatedly.

Energy prices are viewed as a risk by 40% of the US investors surveyed and 55% of the UK investors. They are also viewed as an opportunity by 40% of the US investors and 29% of UK the investors. The polarisation here is obvious, and the country difference matters. The US has a large domestic energy sector, meaning some investors may see higher energy prices as positive for certain holdings. The UK investor base, by contrast, appears more likely to interpret higher energy prices through the household cost-of-living and company-cost channels.

Why some investors think global events do not matter either way

These investors are not necessarily saying that global events do not matter for the world. They are saying global events do not matter enough for their portfolios.

The main reasons both the US and UK investors in our survey cite are diversification, long-term investing, defensive holdings, local exposure, and financial discipline. They also frequently think that portfolio outcomes are driven more by company fundamentals, dividends, valuations, financial advice, local exposure, or long-term compounding, than geopolitical events. In the written responses, this often appears as “markets go up and down,” “over time they smooth out,” “my portfolio is diversified,” or “I invest for the long term.”

There is also a behavioural component. Some investors may choose “no impact” because they do not feel confident predicting global events. Several responses in the ‘no impact’ columns are uncertain, vague, or non-committal. This does not mean respondents believe global events are irrelevant. It may mean they do not know how to connect those events to their specific investments.

The “no impact” camp is therefore divided into two groups. The first is strategically insulated: diversified, long-term, defensive, or locally exposed. The second is cognitively uncertain: unsure, disengaged, or unable to explain the link.

Why some investors think global events will hurt their portfolios

The ‘negative impact’ responses are dominated by concerns about geopolitics. In the US, the most important theme is war and geopolitical conflict, particularly Iran, the Middle East, Ukraine/Russia, and escalation risk. In the UK, negative responses are similar but are even more explicitly tied to the inflationary consequences of global events. They frequently mention energy prices, oil, gas, the cost of living, and the way foreign conflicts can feed into domestic household and business costs.

When investors are negative about global events, their written responses suggest that many see a clear macro chain:

1. Global conflict disrupts energy, trade, or confidence.
2. Energy and commodity prices rise.
3. Inflation and interest-rate pressure increase.
4. Company margins and consumer spending weaken.
5. Stock markets fall and portfolio values decline.

This chain appears again and again in the responses, even when written informally. Investors may not use terms such as “supply shock,” “margin compression,” or “risk premium,” but they are effectively describing those concepts.

The UK-US difference is notable. Our surveyed US investors appear somewhat more willing to see opportunities within global volatility, while the UK investors are more likely to view the same themes through the lens of inflation, energy dependence, and broader economic vulnerability. That may explain why "World Politics and Globalisation cycles" is the highest-ranked cause of portfolio change among UK respondents.

Political uncertainty also has a strong presence in the written responses. In the US, this focusses on President Trump, tariffs, foreign policy, and the perceived unpredictability in government. In the UK, people also mention Trump and US policy, but usually as part of a wider concern about global instability. This implies that investors do not separate domestic politics from geopolitics as cleanly as analysts often do. Leadership, war, tariffs, inflation, and markets are therefore perceived as part of one connected risk cluster.

Why some investors think global events will help their portfolios

The positive camp contains three broad types of investors. First, the optimists who believe technology and growth will dominate. Second, the sector opportunists who own assets that benefit from disruption. Third, the tactical buyers who treat volatility as a chance to buy assets cheaply.

Notably, many investors say they apply a clear investment logic. Essentially, 'disruption creates opportunity'. They cite the effect of global events on an economic recovery, technology, AI, energy transition, lower prices after market sell-offs, defence, oil, gold, emerging markets, and international trade recovery.

The most important positive theme in the US responses is growth opportunity: global events can open new markets, increase demand, create new investment opportunities, and raise returns. Technology and AI are also central. Many respondents expect AI, digital transformation, health care, renewable energy, and innovation-led sectors to do well.

This US-UK divide appears again in the written responses. US positive respondents are frequently enthusiastic about AI, tech, innovation, market growth, and sector opportunities. UK respondents are also positive on technology and innovation, but their positive responses more often include general optimism, economic recovery, and rebounds after conflict.

There is also an opportunistic trading mindset. Some respondents say market dips allow them to buy, or that volatility creates opportunities.

Indirect effects that are implicit in investor logic

The most concerning megatrend in the world today is fiscal debt and deficits – and it is an important part of the broader story about global events. Fiscal issues are less prominent in the written responses from investors compared with war, oil, inflation, and President Trump. Yet the implicit themes run through many identified factors. Indeed, 38% of US and UK respondents view fiscal debt and deficits as a risk. At the same time, around a quarter of people in both countries view them as an opportunity.

So, although investors talk frequently about inflation, rates, taxes, government borrowing, and weaker growth, they do not always connect those concerns explicitly to sovereign deficits. But fiscal anxiety appears indirectly embedded in worries about inflation, interest rates, debt, taxes, and government policy.

Government debt and spending has already prompted portfolio changes by 39% of the US investors and 29% of the UK investors over the last three years. That is lower than technology, world politics, and energy transition, but still significant. It suggests that fiscal concerns are not absent, they are simply less emotionally vivid than war or energy prices.

Thus, investors who are positive on AI may be implicitly assuming that technology can overpower fiscal and geopolitical headwinds. Investors who are negative on global events may be implicitly assuming that debt, conflict, inflation, and policy instability will overpower technological progress. The debate is not simply about whether global events matter. It is about which global force dominates.

Conclusion: The polarisation of opinion on global events means we should brace for volatility and dispersion

In the end, the surveyed investors overwhelming think global events matter because they reach into their portfolio through inflation, oil, politics, growth, and technology. Investors think the world does not matter when they believe their portfolio construction, time horizon, or asset selection can neutralise those forces.

The investor base is therefore not split between people who “understand” global events and people who do not. Rather, it is split between investors who emphasise different time horizons and different asset exposures.

Cultural differences are important. The surveyed US investors are most likely to have changed portfolios because of technology, followed by world politics, while the UK investors are most likely to have changed portfolios because of world politics and globalisation cycles, and then technology.

This suggests that US investors are more likely to frame the future around innovation and technology while UK investors see investing more through the lens of global political cycles.

In investment terms, this means global events should be understood less as a single directional call on equities and more as a source of dispersion. In other words, they just change the relative winners and losers.

Appendix 1

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Deutsche Bank
Research Institute

What a "Strong and Rich Japan" could mean

July 24, 2026

Authors

Mallika Sachdeva
Strategist

Executive summary

- Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. Japan's new economic blueprint seeks to put the country on a path of investment-driven growth to meet the challenges of a new era.
- Japan is not alone in its bid to prioritize industry and rearm, but given a higher starting point for government debt, it will have to find room to spend while maintaining fiscal sustainability. This will inform Japan's approach in very significant ways.
- First, Japan is likely to remain focused on supporting nominal GDP growth, aiming to keep it above government funding costs to create deficit space. Second, Japan will look to actively mobilize domestic savings to fund investments and manage yield pressure. Each of these shifts could have implications for the currency.
- Japan's excess savings have arguably been kept in two places: in cash and abroad. The government is looking to get both moving. Households have around 50% of their $15tn of savings in cash and deposits. Meanwhile, GPIF has 50% of its $1.8tn portfolio abroad. If industry, energy, defence need to come home, so might the capital to fund them.
- Over the past few months, Japanese policy makers have been very focused on stabilizing the currency. However, if fiscal capacity is becoming the most important policy criteria, incentives could shift from FX to yield management.
- The ultimate impact on USD/JPY will depend on the levers the government chooses to pull to manage yields. If GPIF is mandated to bring money back into domestic assets, this could be very bullish for the JPY. However, if the BoJ is coopted to support bonds through renewed JGB purchases, this could be very negative for the JPY.
- Suppressing yield volatility could come alongside bigger moves in FX from here, depending on the approach the government takes. This favours higher JPY vol.

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Important research disclosures located in Appendix 1

Echoes of Meiji

Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. In 1868, pre-modern Japan was staring at the threat of foreign involvement and colonization, having seen the Opium Wars unfold in China. Japan had signed unequal trade treaties that opened Japanese ports to free trade imperialism, prevented Japan from setting tariffs, and exempted foreigners from local laws. Japan's response changed history: under the Meiji Restoration, Japan returned rule to the emperor, strengthened the state, and pursued extremely rapid industrialization.

One could argue that Japan again faces existential challenges given fractures in globalization, intensifying great power competition, insecurity around energy and critical minerals, the need to participate in AI innovation, and external dependence for defence. Japan's answer under PM Takaichi: to return to its roots as an industrial nation driven by state and fiscal policy. The slogan of the 1868 Meiji Restoration was "Fukoku Kyohei" which translates into "Enrich the Country, Strengthen the Armed Forces" or more simply: rich country, strong army. It is perhaps telling that Takaichi's government is introducing an economic blueprint that calls for a "Strong and Rich Japan". (Nikkei Asia).

Invest, invest, invest

Japan's Cabinet has now passed the Honebuto, or the government's new Basic Policy on Economic and Fiscal Management. The document calls for moving beyond "conventional thinking", "putting an end to the trend of underinvestment", and "responding to the era of intense global industrial policy competition". The cornerstone of Japan's proposed policy shift seems to be Y370trillion of public-private partnership investments with the government having identified 17 key strategic areas from AI and quantum, to defence, aviation, shipbuilding, critical minerals, and others. These investments are seen as crucial to boosting potential growth. Indeed, the decline in Japan's potential growth rate has not principally been about demographics as is commonly assumed, but about the collapse in the capital stock (Figure 1). Japan went through decades of very low investment after the 1990 asset bubble burst. It now wants to return to being an investment-driven economy.

Finding the funding: two big shifts

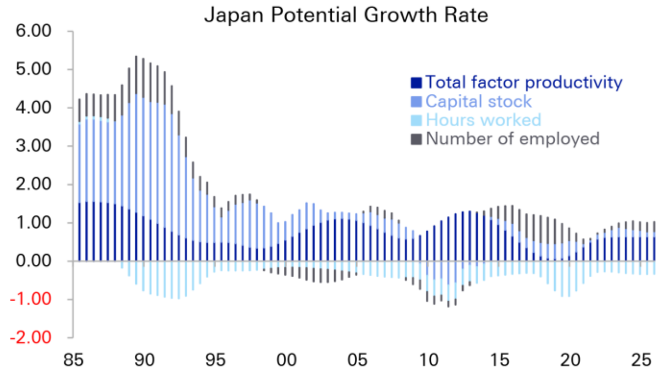
Japan is not alone in its bid to reindustrialize and rearm, but compared to peers, Japan arguably has less fiscal space. Japan's debt to GDP hovers above 200%, while Germany and Korea are under 50%, making it easier for them to pursue significant expansion in their fiscal deficits. While Germany is moving from austerity to expansion, Japan will have to find spending room while continuing to chart a fiscally sustainable path. This will hugely inform Japan's policy approach. There are two important shifts Japan appears to be making to enable fiscal spending.

First, changing the fiscal target. The Honebuto has confirmed a shift in Japan's core fiscal target away from the primary balance. The primary balance will now no longer be a mechanical target; it will be managed over many years, and be allowed to deteriorate temporarily as required. The new fiscal target is now the "stable reduction of the total debt-to-GDP ratio for national and local governments." The second notable and enabling shift is the pursuit of an "asset management nation" strategy. Japan is looking to more actively mobilize domestic savings to fund investments and help keep government financing costs manageable. Each of these shifts could have significant long-term implications for the currency.

The new fiscal approach means keeping r below g

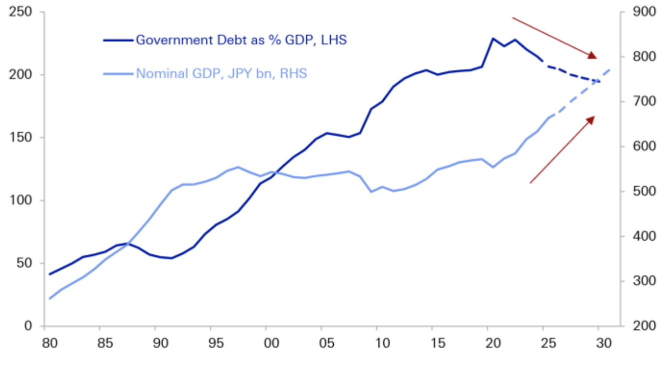
We think debt sustainability arithmetic has become key to understanding Japan's economic policy. Recall that: $\Delta d_t \approx (r-g)d_{t-1} + pd_t$ where d is the debt-to-GDP ratio, pd is the annual primary deficit, g is the nominal GDP growth rate, and r is the average interest rate paid on outstanding debt. The formula implies that Japan can run primary deficits and still achieve a declining debt-to-GDP ratio, as long as r is below g. In other words, Japan's cost of funding needs to be below nominal GDP growth for Japan to have fiscal deficit space. One might consider Abenomics' bid to return Japan to inflation as crucial to putting it on a path to fiscal health. As Figure 2 illustrates, the peaking in Japan's debt-GDP ratios has gone hand-in-hand with the revival in Japan's nominal GDP.

Figure 1: Japanese potential growth has declined on decades of underinvestment – this could now change



Source: Deutsche Bank Research, Bank of Japan

Figure 2: Rising nominal GDP has helped bring down debt-to-GDP



Source: Deutsche Bank Research, Haver Analytics

The extent of fiscal deficit space will crucially depend on the interplay between r and g with Figure 3 illustrating the potential space created by different (r-g) scenarios. We note that the Honebuto outlined a 3% nominal GDP target (g) for Japan till 2040, made up of a 2% inflation target, and 1% real growth target, with aims of increasing it. This perhaps sheds light on why the government has become sensitive to the 3% level on 10Y yields. To be sure, Japan's overall cost of funding remains much lower for now at near 1% given just a fraction of debt is refinanced every year, but there are points of vulnerability. While the weighted average maturity of debt outstanding is around 9 years, BoJ holds close to 50% of outstanding JGBs against which they pay interest on excess reserves to banks. This means that almost half the consolidated liability base is in fact linked to front-end floating rates. Therefore, not only are long-run government bond yields above 3% not likely to be sustainable, front-end rate hikes also contribute to consolidated fiscal costs. Very roughly, every 100bps of rate hikes costs Y5tn or around 0.7% of GDP.

The (r-g) formula not only explains the importance of understanding the drivers of 'r', but also explains the sensitivity to 'g'. It helps one appreciate why policymakers remain so worried about upsetting the balance on hard-won nominal GDP growth. Indeed, Japan tends to react very cautiously and dovishly to any signs of growth risks (tariffs, Iran war) and has preferred to let inflation run hot and durable before tightening.

Japanese policy makers face a fine balancing act in optimizing this formula: dovish monetary policy may help to support g, but is also driving up r via inflation expectations reflected in long-run bond yields. Conversely, hawkish monetary policy could hurt 'g' and also raise the cost of 'r' via reserve payments. In light of this, the focus on trying to manage long-end yields makes more sense.

How might Japan do this? Changes to supply-side of debt management have been tried via reduced long-end issuance; the government may now be trying to more actively influence demand.

An Asset Management Nation

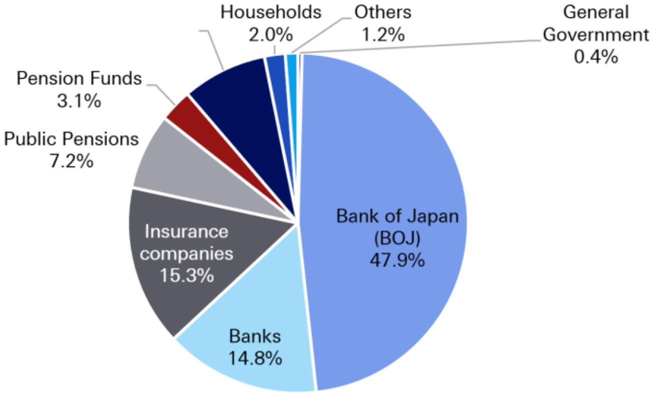
Very simply: Japan needs to find more buyers of JGBs (Figure 4). It is natural to look at where its own domestic capital has been invested. The answer points to two big places: in cash and overseas. Japan has close to USD5tn in portfolio assets held abroad across central bank, household, pension fund, insurance, and bank balance sheets, more than the size of GDP. Indeed, our research has long shown that one of the best metrics of fiscal space is a country's external position, and Japan may need to draw on it. Intuitively, the bid for strategic autonomy around the world should coincide with a growing "nationalism of capital." If industry, energy, defence all need to come home, so might the capital to fund it. We have explored this in earlier research here. More of Japan's money could now be needed at home. Finance Minister Katayama has been explicit that Japan's ambitions to create a growth-oriented economy will coincide with a "strong stock market and positive interest rates" in which domestic citizens should reap benefits. The other perspective is they will help provide the money.

Figure 3: Keep g above r is key to Japan's fiscal space. Japan does not want to return to the lost decades

Scenario	Nominal GDP	Nominal Rates	Debt Stabilizing Primary Balance	Scenario Description
1	4.5%	1.75%	**-5.5%**	*2025 Regime*
2	3.0%	2.0%	**-2.2%**	*Japan manages rise in yields to keep them below g*
3	0.0%	1.5%	**1.5%**	*Return to Lost Decades (1995-2010)*

Source: Deutsche Bank Research, Bloomberg Finance LP

Figure 4: Japan will need to find more buyers of JGBs



Source: Deutsche Bank Research, Ministry of Finance

Where the money could come from

Pension fund and household assets appear to be the two big areas of focus for markets. The government pension fund, GPIF, currently has 50% of its $1.8tn portfolio abroad. Meanwhile, households have around 50% of their $15tn (!) of savings in cash and deposits (Figure 5).

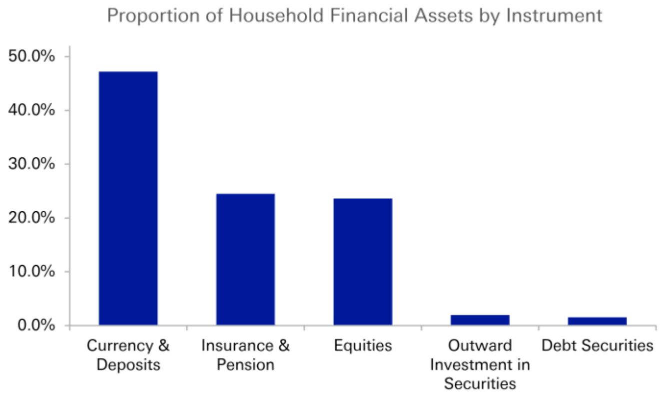
Movement of household deposits into JGBs could be encouraged through a few channels. JGBs may be brought under the NISA tax-exempt investment scheme; the retail JGB offering may be expanded or made more attractive. There were also references in the Honebuto to advancing on-chain finance, with the power of tokenization potentially being tapped to encourage financialization of savings. Mobilizing household savings could be material for yield management given their sheer size. For context, the 1000tn yen of household deposits are almost the size of the entire JGB market. It is not however likely to have any FX impact as it would mostly be an internal domestic flow.

In contrast, a shift from GPIF to invest more in domestic assets could be *very* material for FX, with PM Takaichi noting this is a key initiative. GPIF's current policy mix prescribes 25% investment in domestic equities and debt respectively, with +/-6% bands around it. If GPIF simply shifted to the top end of current permissible ranges from current levels (as of March 2026), this could bring around $200bn back to Japan across domestic equities ($130bn) and domestic bonds ($75bn).

The bigger shift to play for however is an early policy review of GPIF's overall asset mix that revises up the domestic asset targets themselves. In Figure 6 we illustrate scope for inflows at different levels of target allocation to domestic bonds alone. Hypothetically, if the target share of domestic bonds doubled to 50%, over USD400bn could flow into JGBs from GPIF. While this may seem extreme before Abenomics in 2012, GPIF had a 67% target allocation to domestic bonds in their policy mix. One could reasonably expect any flows to occur with a glide path, but the scale alone could be significant enough to impact the currency.

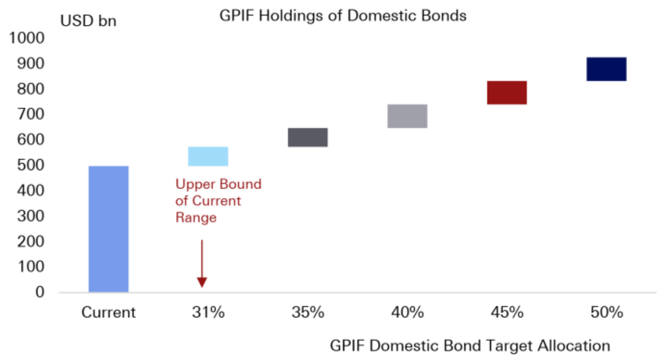
We note that GPIF has a current return objective of a 1.9% real return above nominal wage growth. The extent of any domestic reallocation will depend on how much Japan's domestic assets can deliver that return, or whether government pension returns are made subservient to financing costs for now. Another group that may be called upon to play a bond absorption role could be local banks. This is a source of financial repression has been used extensively around the world and could return where deficit financing demands it.

Figure 5: Japanese households hold half their savings in cash and deposits



Source: Deutsche Bank Research, Haver Analytics

Figure 6: GPIF could be encouraged to bring more money back into domestic bonds



Source: Deutsche Bank Research, GPIF

A shift to yield management could lead to bigger FX moves

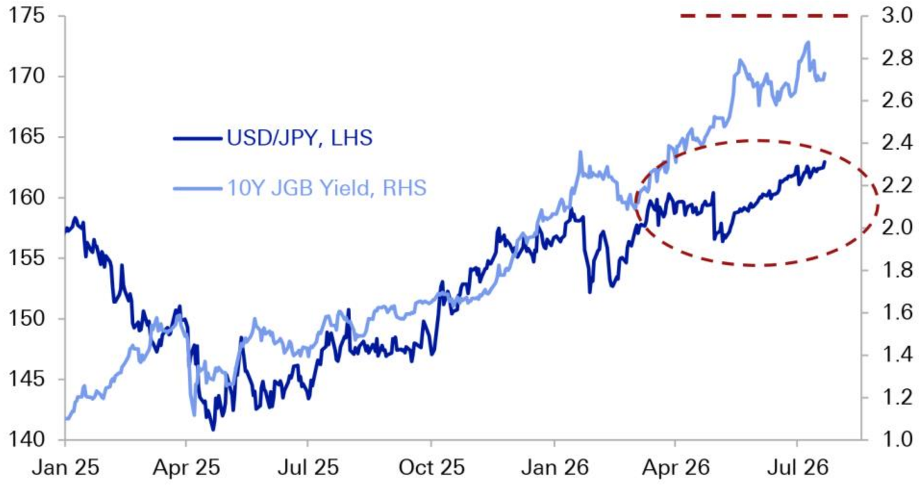
Over the past few months, Japanese policy makers have arguably been solving for stabilizing the currency. Household angst and imported inflation risks have likely encouraged this. Policymakers have engaged in verbal, rate check and actual intervention this year. And, Japan has in fact been successful in keeping USD/JPY near 160 amid the US-Iran war, a major global energy shock, and volatility in Fed pricing.

If, however, fiscal capacity is becoming the biggest and more existential policy criteria for Japan, then incentives may be shifting from FX management to yield management: from capping USD/JPY to capping 10Y yields and borrowing costs.

Suppressing interest rate volatility may come at the cost of amplifying it in FX, depending on the specific levers that the government chooses to pull. If GPIF is mandated to bring a larger share of their assets back into domestic assets, this would be very bullish for the JPY. We have previously identified GPIF as the biggest weapon in Japan's external arsenal. Equally, if BoJ is encouraged to accelerate tightening to manage inflation risk premium in long-end yields this too could be bullish JPY. If, however, the BoJ is coopted to support yield management through a more active role in the bond market, potentially resuming JGB purchases, this would be negative for the JPY. Similarly, if BoJ is encouraged to keep monetary policy dovish to prioritize (r-g) considerations this too would be bearish JPY.

Ultimately, the direction of travel for USD/JPY will depend on which measures the government favours. Clearer and more imminent signs of GPIF repatriation would favour the stronger yen tail, while pressure on BoJ to manage rates and yields again could open the weaker tail. Efforts to suppress yield volatility could come alongside bigger moves in FX from here. This favours higher JPY vols. We note USD/JPY vols are near multi-year lows with a 1Y USD/JPY straddle breaking even roughly below 150 or around 170.

Figure 7: Policy focus could be shifting from capping FX to capping yields



Source: Deutsche Bank Research, Bloomberg Finance LP

Appendix 1

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Deutsche Bank
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Time billionaires want more risk. Is AI to blame?

July 22, 2026

Authors

Luke Templeman
Research Analyst

Galina Pozdnyakova
Research Analyst

Executive summary

Are you a time billionaire? Some say "yes" if they expect to have more than 1bn seconds (or 31.5 years) in front of them. And as the number of younger investors increases, and their time horizons span 30, 40, or 50 years, they want two things: 1) More risk than they have today, 2) AI to manage their money and trade for them.

So, is AI convincing investors, particularly young people, to increase the current level of their investment risk? In this report, we use proprietary data from our dbDataInsights team to break down exactly who wants more investment risk and over what timeframe. We highlight that both Americans and Britons under the age of 55 are particularly keen to increase (relative to today) both their risk tolerance and their dependence on AI to manage their portfolio. The younger the age group, the more they say they will increase their risk over the coming 1, 3, and 5 years.

We then discuss various factors that are contributing to the increased desire for both risk and AI advice. Behavioural psychology feeds directly into the equation, particularly as 'social' investing has become a trend. 'Aspiration inflation' is also very tangible but we point out that young investors (and their risk appetite) are the product of the environment that has been created for them. In the end, there are pros and cons.

Several conflicts and conclusions come out of our work. The first is that it is intriguing that consumer sentiment is so low at the same time that the average person wants to take on more investment risk. A second is that this desire for risk at a household level comes as experts increasingly point to intensifying global systemic risks. A third is that the willingness to turn to AI for investment advice comes as the socials winds against AI more broadly are growing in strength.

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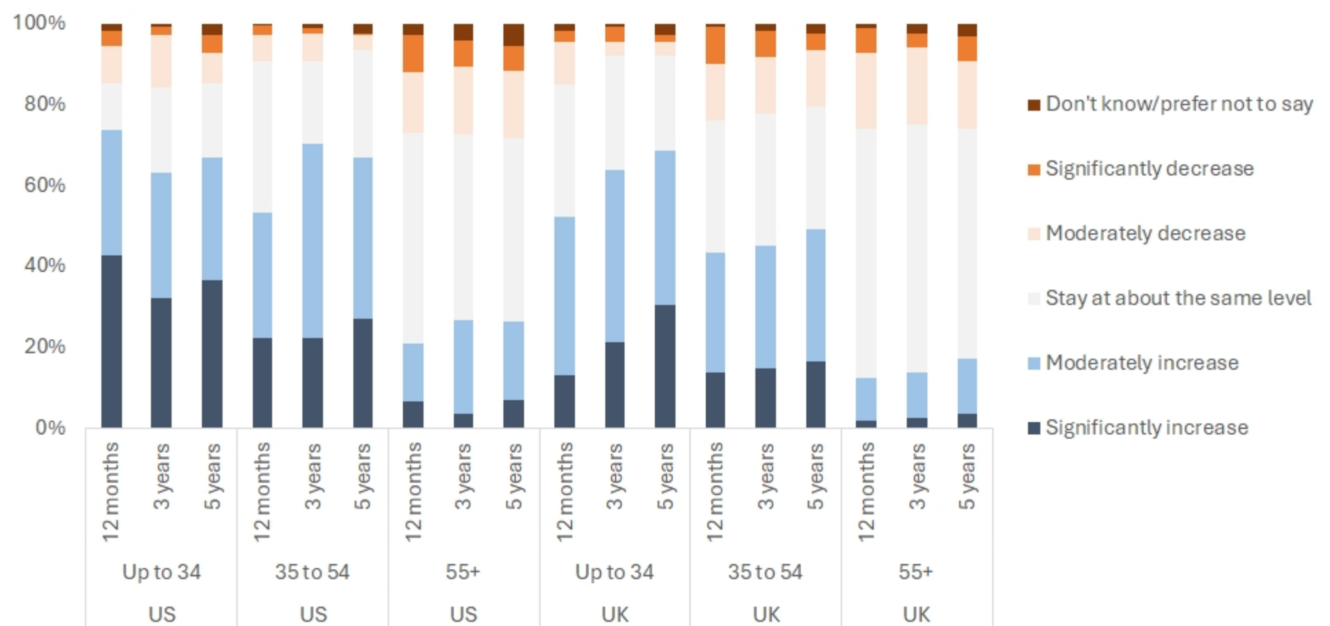
An appetite for risk

At one level, it seems strange that six years into a decade that has been so volatile, people want to take more, not less, risk with their investments. Indeed, the following chart shows that all age groups (up to the 55+ group) want to substantially increase their investment risk over a 1-, 3-, and 5-year time frame.

Figure 1: Over each of the following timeframes, do you expect your level of risk-taking in relation to your main investment portfolio to...



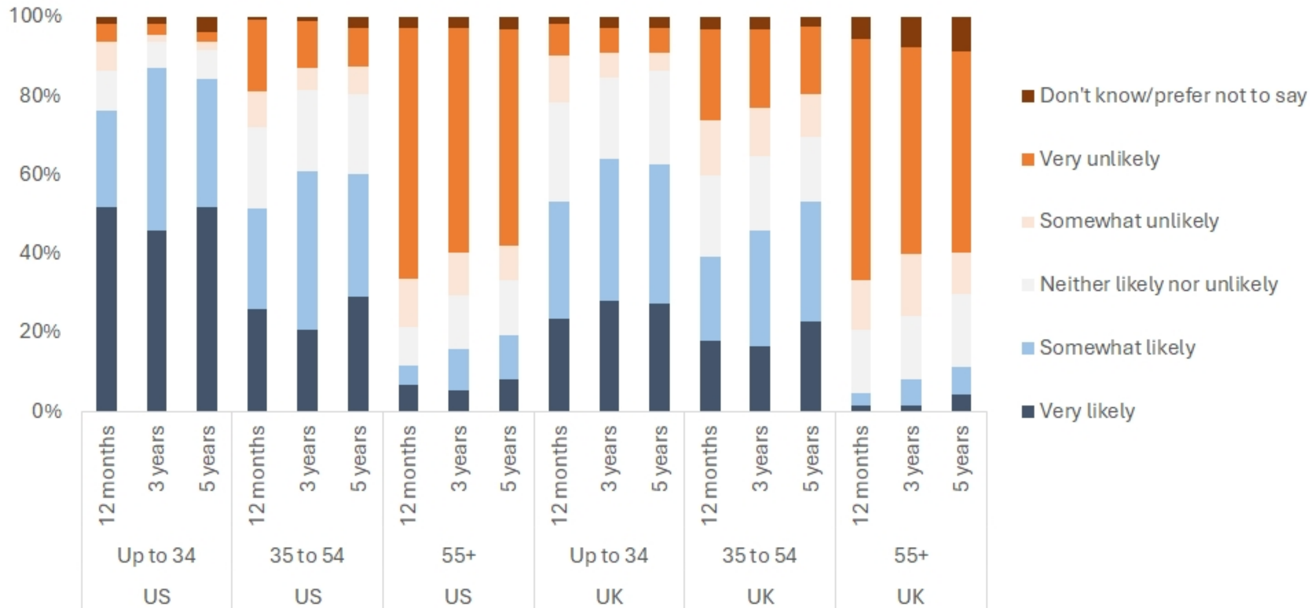
Source : dbDataInsights

A stark observation is the age gradient. In short, the younger you are, the more likely you are to want to take even more investment risk in the future. The effect is most pronounced in the US but the trend is the same in the UK.

Is AI encouraging risk?

People are not only using AI to find investment advice, but also their shift towards it is accelerating. As the following chart shows, people under the age of 55 expect to show a strong desire to move to an AI managed investment method over the next 1, 3, and 5 years. This phenomenon is particularly aggressive in the US but still pronounced in the UK.

Figure 2: Within the following timeframes, what is the likelihood you will switch from your current investment management method (self-managed or human advisor) to a Generative AI-managed solution?

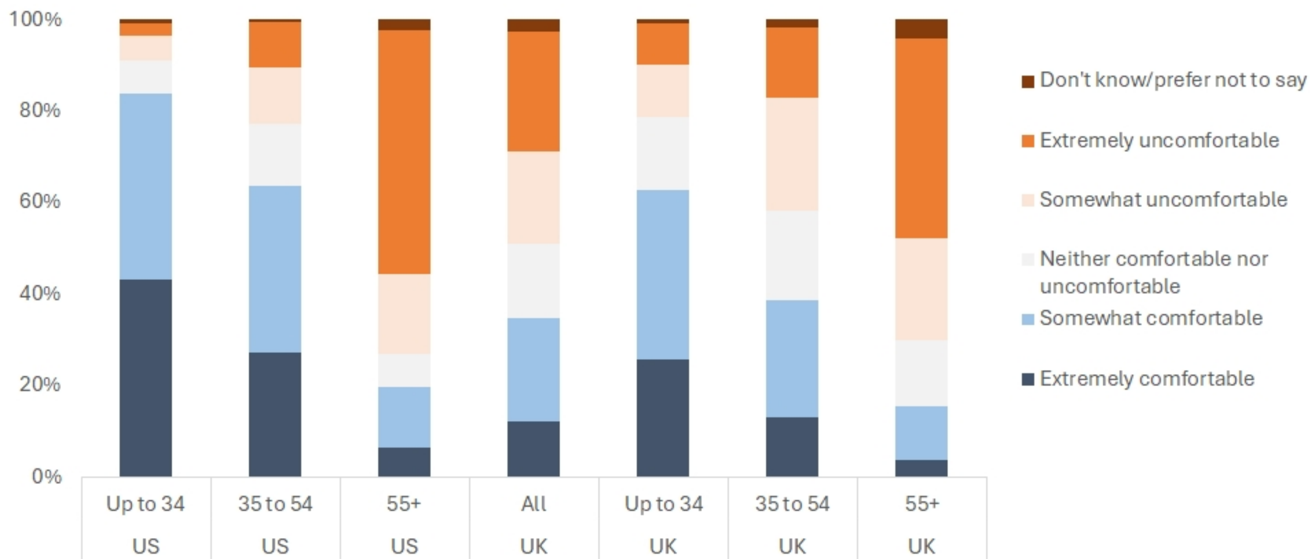


Source : dbDataInsights

Will AI trade for you?

Regardless of how AI is or is not encouraging risk, people are certainly increasingly comfortable with AI automatically making trades in their investment account. For example, Americans up to 34 years old are happy giving AI responsibility for automatically executing trades on their account by a ratio of 10:1. In the UK, the same age group is happy with AI by a ratio of 3:1. The ratio falls as people age.

Figure 3: How comfortable would you be with a generative AI automatically executing investment trades on your behalf, if you had set the parameters (e.g., buy/sell limits, asset allocation)?



Source : dbDataInsights

Does AI give confidence to time billionaires?

AI may amplify risk appetite, particularly for young people, because it reduces the friction around decision-making. Everyday investors can now ask an AI tool to explain asset allocation, compare funds, summarise macro risks or model hypothetical scenarios. Even if the output is imperfect, the interaction can create a sense of fluency. That matters because people are more likely to take action when a complex problem appears legible.

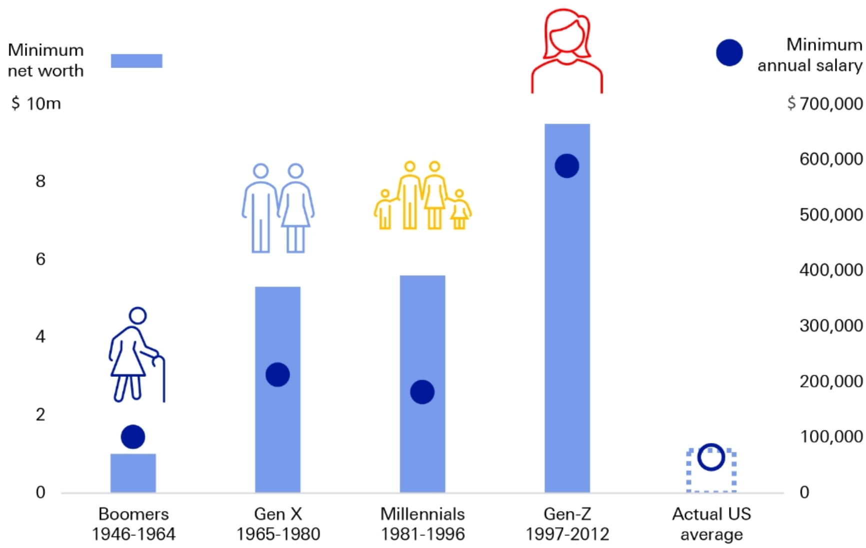
Some research supports this direction of travel. The World Economic Forum has previously highlighted that Gen-Z and millennial investors are embracing AI and digital platforms for financial advice and education, and that younger investors place greater importance on intuitive, technology-enabled wealth-management systems. The WEF has noted that 45% of Gen-Z and millennials began investing in early adulthood, compared with 15% of Gen X and baby boomers.

What else is driving greater risk taking?

Aspiration inflation

Some theories suggest the ease of 'aspirational connections' and 'dream scrolling' on social media has fuelled sky-high expectations for Gen-Z. In fact, one survey shows that Gen-Z thinks they need twice as much wealth and triple the annual salary as the previous generation to achieve "financial success".

Figure 4: Survey: "What does financial success look like?“



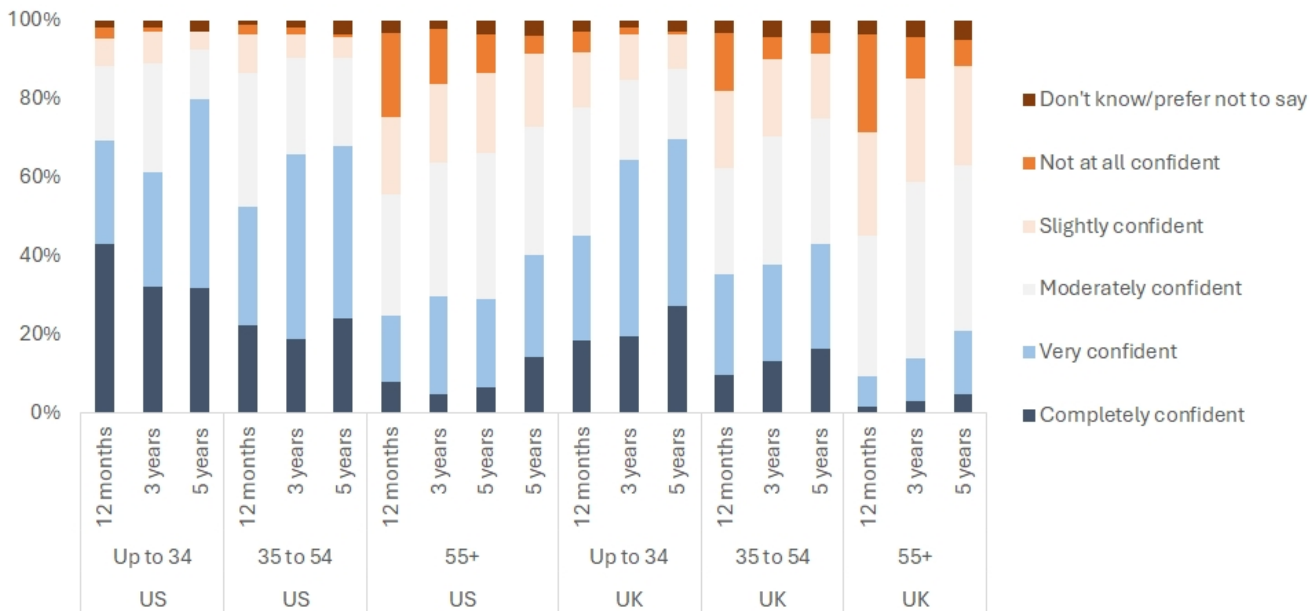
Source : Empower "Secret to Success" study: [Secret to Success](#) Empower, November 2004, Deutsche Bank. (Note: Average salary as of 2024, average net worth as of 2022)

Of course, the flip side of the same coin is that younger people are increasingly worried about housing affordability, education costs, uncertain retirement prospects and perceptions of stagnant real income growth. As a result, this may encourage them to see traditional saving as insufficient. If higher-risk investments can appear less like speculation and more like necessity then risk-taking is not a rejection of prudence, it is an attempt to close an opportunity gap.

Bullish return expectations

The chart below shows that people in all age groups are confident that their investment returns will meet or exceed their expectations. Again, age groups matter. Younger people are overwhelming confident of strong returns across 1, 3, and 5 years.

Figure 5: Over the following time frames, how confident are your that your main investment portfolio will meet or exceed your expectations?



Source : dbDataInsights

Rallying against the risk of doing nothing

Portfolio risk appetite is not inversely related to global risk perception. In a risky world, households may feel that the cost of remaining conservative is also rising. If inflation, technological disruption and asset-price dispersion are perceived as threats, risk-taking can become a hedge against being left behind.

Young investors are a product of their environment

Many younger investors have been socialised as they came of age in a market culture shaped by mobile trading, social media, crypto assets, zero-commission brokerage, online financial influencers and rapid cycles of boom-and-bust narratives. Investment decisions are discussed, compared, gamified and sometimes celebrated in digital communities. This can make risk-taking feel less isolated and less exceptional. In turn, this amplifies psychological issues of social proof, herding, and fear of missing out. In the end, the emotional cost of failure is lower.

Anthropological work on risk argues that societies do not merely discover risks; they classify, narrate and prioritise them. Mary Douglas's cultural theory of risk argued that communities highlight different hazards depending on their social organisation, values and forms of trust. What counts as a salient risk is, in part, culturally produced.

Studies also emphasise that risk-taking is often mediated by rituals, expertise and tools that make uncertainty feel manageable. In modern finance, models, apps, dashboards and AI systems can serve a comparable psychological function. They do not eliminate uncertainty, but they make uncertainty appear structured and therefore more actionable.

The increased ease at which people can access risky products raises the idea of 'prospect theory'. This highlights that people become risk-seeking when they perceive themselves to be in the domain of losses. They then can be more likely to gamble to return to their reference point rather than accepting a smaller certain loss. An investor who is underwater relative to a recent peak, or who feels they have 'missed out' relative to peers who bought earlier, is, under this framework, more likely to reach for a higher-variance trade to close that gap.

Other psychologists have highlighted how sensation seeking can correlate with risk-taking in gambling, extreme sports, and financial speculation. It tends to be higher in younger populations — a pattern consistent with our survey findings.

Do consequences feel lower than in the past?

Risk-taking is not simply a function of appetite; it is also a function of perceived downside protection. Behavioural researchers analyse how humans recalibrate their risk-taking in response to changes — real or perceived — in the safety of their environment.

Consider the Peltzman effect. This describes how people can take greater risks if they know they are protected from downside risk. Evidence for this has been argued by researchers in various domains:

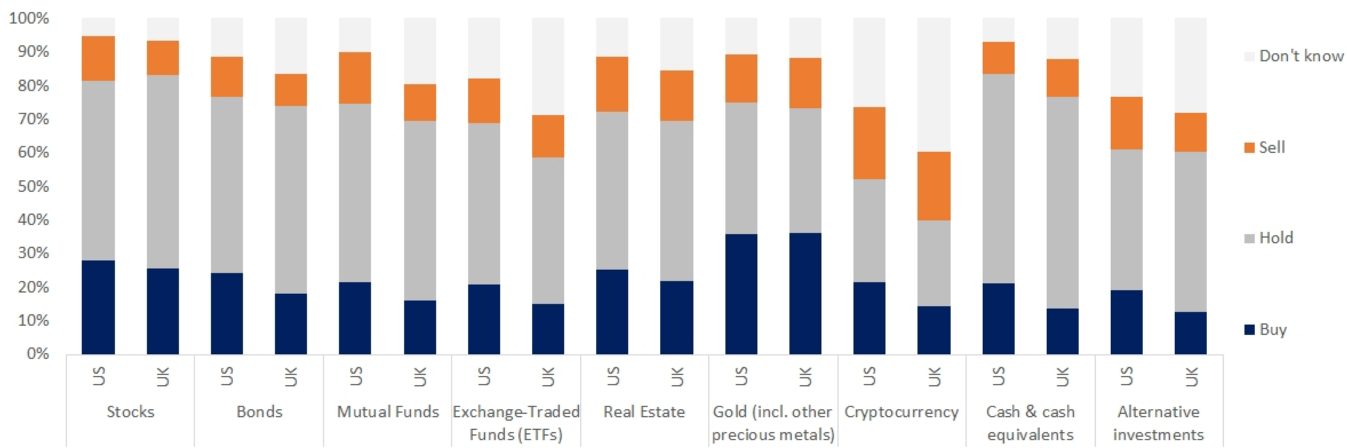
- Some drivers were found to be involved in more accidents when their cars had greater safety features
- The introduction of HANS safety devices in Nascar was argued by some to be related to an increase in more aggressive racing
- Skydiving/wingsuit users readily admit that the increase of the quality of equipment has made them push the limits of the sports

Application to finance and investing

Post-2008, investors have become used to policymakers increasing their response to shock events. At the same time, financial innovation has created the perception in some that risk is either lower or can be managed better. Consider that every major US market decline since the 2008 financial crisis has been followed by a relatively timely recovery to new highs. A generation of investors has now only ever experienced dips as buying opportunities – a very different experience from the multi-year drawdowns and stagnant markets of the 1970s or the 2000s.

Hence, the number of people who say they treat uncertainty as an opportunity to take on traditional risk assets. The following chart shows that people think that going into traditional 'risk' assets is the best strategy during periods of uncertainty. In fact, the average American wants to buy, rather than sell, equities during uncertain periods by a ratio of 2:1. In the UK that ratio rises to 2.5:1.

Figure 6: In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?



Source : dbDataInsights

Of course, investors may agree that the world is risky while also believing that AI, technology stocks, private markets, crypto assets or thematic funds offer the best path to future security.

In addition to general faith in policymakers, several other factors may be encouraging risk:

- Deposit insurance, broker-dealer protections, and post-2008 regulatory reforms have reduced certain tail risks that were live concerns a generation ago, such as a brokerage or bank outright failing with a customer's assets inside it.
- Diversified, low-cost index products have made it structurally harder to lose everything through a single-stock mistake compared with earlier eras of concentrated, high-fee retail investing.
- Information is more abundant and more immediate, which may create a (not necessarily accurate) sense that fewer unknowns remain and that outcomes are more predictable.
- The practical consequences of experimentation have been reduced by fractional shares, low-cost funds, commission-free trading and the ability to start with small sums. A young investor can now test higher-risk assets without the same minimum account sizes or advice costs that once limited participation.

Implications for wealth managers and policymakers

Younger investors are unlikely to be persuaded by generic warnings about risk. As such, AI should be framed as a support mechanism rather than a substitute for suitability, behavioural coaching and long-term planning.

For policymakers, the issue is not whether AI-driven investment engagement is good or bad. It is how to ensure that increased access is matched by adequate disclosure, financial literacy and safeguards against misleading personalisation. If AI makes investing easier, it should also make risk more understandable.

If the current environment of ever greater risk taking continues, there are several key concerns that may exacerbate the downside risks:

- Leverage positioning: US margin debt has pushed to over 4.5% of GDP. That is well above prior peaks in 2021, 2007 and 2000. Retail investors can easily access leverage through brokers, ETFs, or other specialised products.
- Speculation has broadened: New avenues for risk taking include the ease of options trading (particularly same-day and binary options), leveraged ETFs, and meme assets, prediction markets and sports gambling.
- Safety buffers are shrinking: Household savings rates are falling in the US and other key countries

Conclusion

The common thread across not only financial decisions, but also auto safety, extreme sports, medicine, and gambling is that humans do not hold risk-taking constant when their environment changes. They recalibrate around a target level of risk, and that target appears to rise when protective factors (real or perceived) are introduced. Applied to investing, this suggests the relevant question is not just, ‘Why are people taking more risk?’ but, ‘What changed in their perceived environment that permitted it?’

AI may be one driver of this desire for risk but there are others, too. Yet, it is concerning that perceived consequences may have declined faster than the actual potential consequences. AI can widen this gap if it gives users the impression that risks are fully understood or controllable. Our data points to pros and cons: AI may democratise investment engagement, but it may also compress the distance between curiosity and action.

The younger cohorts in our survey may have greater time horizons, but they also face housing, employment and debt pressures that can constrain their ability to absorb volatility. The next generation of advice must therefore be both more digital and more behavioural.

Appendix 1

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Open-source AI 101: the battle for the future of AI

July 21, 2026

Authors

Adrian Cox
Research Analyst

Open-source versus proprietary AI models in a nutshell

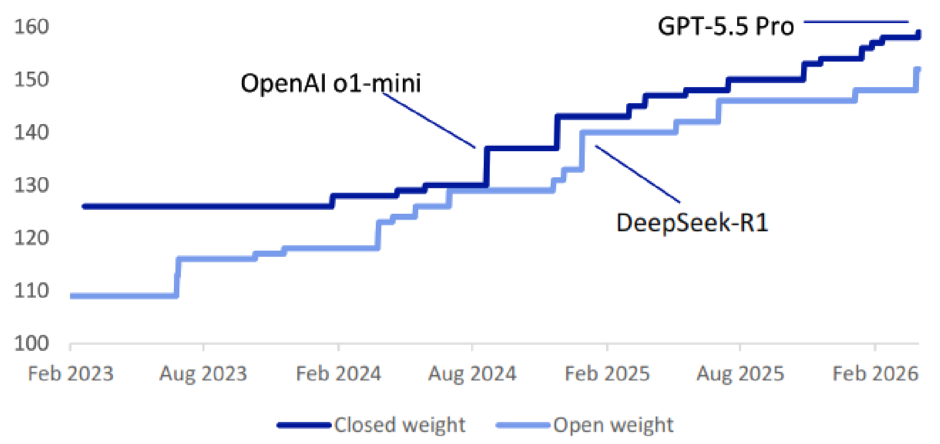
AI-related stocks tumbled at the end of last week on a second DeepSeek moment, as the latest Chinese open-weight AI model reminded investors that cheap, light alternatives are snapping at the heels of the state-of-the-art proprietary models.

In this brief report for generalists, we explain this latest technology format war:

1. Open-what? A new twist on age-old format wars
2. What are open AI models?
3. Why are they disrupting the markets?
4. Why would a company give its model away?
5. What comes next?

The emergence of DeepSeek's breakthrough R1 model in January last year wiped $600bn off the market value of Nvidia in one day on concerns about whether vast hyperscaler capital expenditure was justified in the light of a "good enough" model developed in China using older chips and trained for a fraction of the cost.

Figure 1: Epoch Capabilities Index score: open-weight models narrow the gap with closed-weight proprietary models (data to end April 2026)



Source : Epoch AI, Deutsche Bank Research

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A similar release on Friday revived those concerns, with the Magnificent Seven (mostly hyperscaler) stocks shedding 1.8 percent and the Philadelphia Stock Exchange Semiconductor Index turning down 1.6 percent, bringing its loss for the week to more than 10 percent. Alphabet's Google, Microsoft, Amazon's AWS, Meta and Oracle are set to spend about $700bn building AI capacity this year, up by about 70 percent from last year.

Even so, while powerful open models raise questions about the business models of proprietary model makers and semiconductor demand, they may be bullish for AI adoption, applications and enterprise customisation. Greater competition, more efficient models and lower prices may supercharge a new phase of growth.

1. Open-what? A new twist on age-old format wars

Open-source software is nothing new.

- A study by Harvard Business School in 2024 found that about 96 percent of commercial programs include some code created or distributed for free by public-facing tech forums. Without it, enterprises would pay 3.5 times more, or roughly $8.8 trillion, to build their software and platforms, it found.

But open AI models are contentious.

- Some critics saying that once released they may be less secure and more likely to be misused. (We mainly use the term "open models" for readability and explain the distinction between open-source and open-weight in the definition section.)
- Some open models have also been accused of being trained through "distillation", using the outputs of more capable proprietary models to teach a new model, which may breach providers' terms and magnify errors, biases or safety weaknesses in the original models. Critics also wonder out loud about whether some open models have been geared more towards scoring highly on benchmarks – so-called benchmaxxing – than for solving real-world problems.

Meanwhile, concerns have been rising about whether proprietary models can be depended on.

- Anthropic's most powerful Claude models were recently de facto banned for several weeks on security concerns. Critics also say calls by proprietary model makers for safety regulation would in fact protect incumbents.

The debate is reminiscent of past format battles.

- Every great technology needs a rival. In the "war of the currents" of the late 1880s, Thomas Edison defended his direct-current (DC) empire by portraying the alternating-current (AC) system promoted by George Westinghouse and Nikola Tesla as lethally dangerous—even supporting demonstrations in which animals were electrocuted with AC.
- And remember VHS versus Betamax video cassette players in the late 1970s and early 1980s? VHS did not win by being the better technology: JVC licensed it broadly, manufacturers produced cheaper machines, recording times were longer and a self-reinforcing ecosystem developed around it. Open-source AI could follow the same path. It does not need to beat every proprietary model on every benchmark. It needs to be good enough, inexpensive and ubiquitous.

However, while the contest between open and proprietary AI is often presented as a winner-takes-all format war, the most likely outcome is a combination of the two.

- At the consumer level, it may resemble Apple and Google's Android, a controlled premium ecosystem coexisting with a larger, more customisable ecosystem. And at the industry level, it may resemble the IBM personal computer, where widespread compatibility commoditised the foundational technology while shifting significant value into chips, software, services and applications built around it. Many organisations will use both.

2. What are open AI models?

There are three distribution models. For users, it boils down to control and – to coin a phrase – with greater control comes greater responsibility.

Proprietary, or closed, models are generally the plug and play option, accessed through the developer's application, an application programming interface (API), or a managed cloud services.

- For a rough analogy, think of Apple's iOS operating system or Microsoft's Windows, where the company controls the underlying technology and the terms on which others can use it.
- With proprietary AI models, the developer keeps the model weights, training methods and most details of the training process private. It controls pricing, updates and safety restrictions, and can alter or withdraw access. Prominent examples include OpenAI's flagship GPT models and Anthropic's Claude models.

Open-weight models are open-source-lite.

- Their trained parameters – the billions or trillions of numerical values that encode much of the model's learnt behaviour – are typically released for users to download, subject to an applicable licence. You can run the model on your own infrastructure, adapt or fine-tune it, and technically alter or remove its safeguards.
- But the release may not include the original training data, the complete training code or enough information for you to reproduce the model from scratch. Many of the models referred to casually as open-source – including DeepSeek's R1 and Meta's Llama models, as well as many released by France's Mistral – are in fact more accurately described as open-weight.

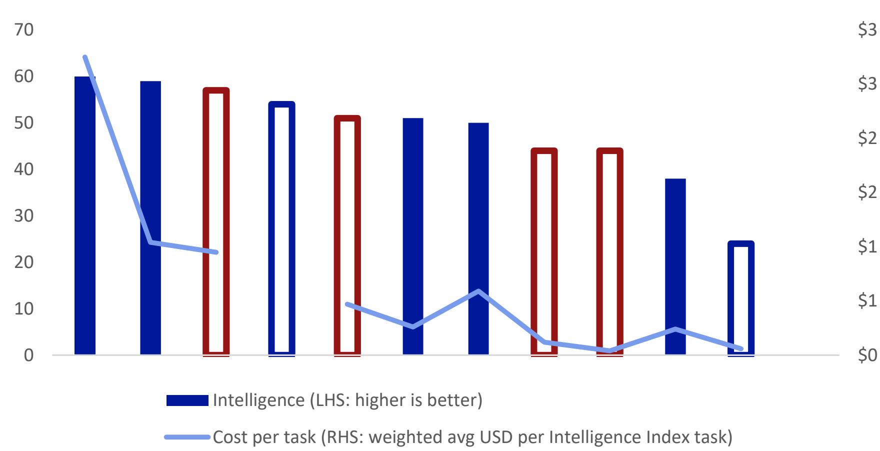
Genuinely open-source AI systems are much rarer.

- Under the Open Source Initiative, they should give users the freedom to use, study, modify and redistribute the system, supported by access to its parameters, training and inference code and detailed information about the training data. The grand-daddy of open-source is Linux, a free operating system created in 1991 and found in nearly all cloud infrastructure and the majority of the world's servers and mobile devices.

Consequently, open-source models often offer:

1. **Lower costs, particularly at scale or for specialised tasks.** If you run the model yourself, you can avoid the developer's per-token API charges and can select smaller or compressed models requiring fewer calculations. **The downside?** Downloadable does not mean free: you still have to pay for compute, electricity, engineering, monitoring and maintenance. Open models may also not be as universally capable as the leading proprietary systems and may lack their polished integration with wider products and services.
2. **Greater customisation and control.** You can fine-tune a model for your industry, language or task, rather than relying on a one-size-fits-all service. **The downside?** Although open models are cheaper over the long-run, you need to invest a lot upfront in infrastructure and expertise. You have to take more responsibility for testing, guardrails, cybersecurity, updates and reliability—though managed cloud providers can fill some of the gaps.
3. **Local deployment.** You can install some models on your own servers, or even on "edge devices" such as cars and phones, rather than relying on a connection to the cloud network run by the provider. That makes them potentially faster (less "latency") and more secure as you don't need to share your data. Once you download it, it's yours, and API access cannot be suspended without warning. That is key for some regulated organisations where data can't leave the premises. **The downside?** Not every model can run economically on local hardware, and responsibility for maintaining performance and protecting the system falls on your shoulders.

Figure 2: Intelligence index of selected AI models vs cost per tasks: US (blue) and proprietary (solid colour) models are no longer far ahead of Chinese (red) and open-weight (white with border) models



Source : Artificial Analysis Intelligence Index, Deutsche Bank Research. Data as of July 20, 2026

3. Why are they disrupting the markets?

Open models challenge four assumptions about proprietary models embedded in AI valuations.

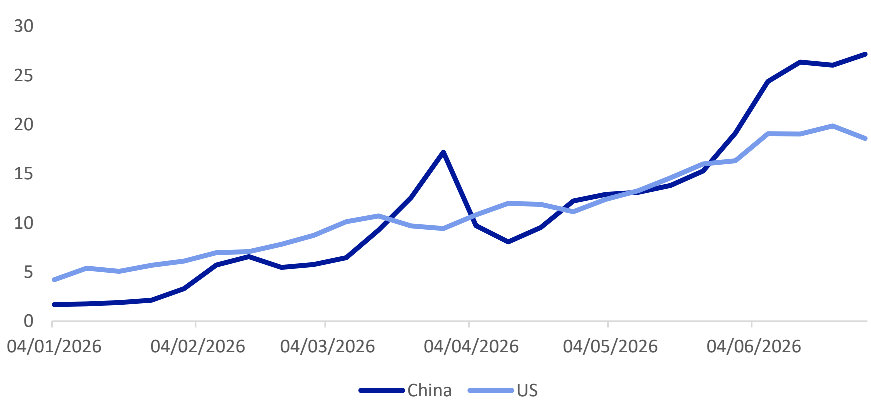
First, scarcity. If capable models can be copied, modified and hosted by many providers, AI models might begin to look like a commoditised software infrastructure rather than a uniquely monetisable product.

Secondly, pricing power. AI agents and "tokenmaxxing" employees are guzzling up vast numbers of tokens, the small units of data that AI models process. As pilots scale up and AI subsidies come to an end, the leading makers of comprehensive models are starting to charge for each token rather than a flat rate. For cost-conscious companies, open models that are good enough for most purposes are looking increasingly attractive.

Thirdly, capital investment. If you can get something like the same performance out of an open model with lighter training and inference (running) horsepower, can the projected spending on chips, networking equipment, electricity and data centres be justified?

Fourth, US technological moat. Chinese open models weaken the perceived durability of US supremacy. The Trump administration says it is in "a race to achieve global dominance in AI" and has driven policies to strengthen domestic production and adoption and stifle Chinese access to cutting-edge chips and other technology. Chinese models have overtaken US models this year in terms of the number of tokens processed on each by the OpenRouter developer platform.

Figure 3: OpenRouter traffic: trillion tokens going through model platform this year, broken down by country of model



Source : OpenRouter, Deutsche Bank Research

For more, see our recent report "AI IPOs 101: why they'll change the boom forever".

4. Why would a company give its model away?

Open models may be commercially viable because producers do not need to monetise the weights themselves. It's about losing the battle to win the war.

1. Monetising complementary products

- Developers can charge for convenient hosted APIs, premium subscriptions, faster inference, enterprise support, fine-tuning, security and guaranteed service levels. The equivalent of in-app purchases for a free app gives less tech-engaged customers the best of both worlds.

2. Building an ecosystem and standard

- A widely adopted model attracts developers, tools, optimisations, fine-tuned variants and research. All of that can improve the model, lower the cost of acquiring new customers, and make its interface an industry standard.

3. Commoditising competitors' profit pool

- This is a classic case of commoditising the complement: giving away something for free (or very little) to stop your rival from earning excess returns, while increasing demand for (and monetising) your own surrounding infrastructure. So cloud providers that produce or encourage open models can benefit from selling compute to host open models; consumer platforms can benefit from better recommendations and advertising; and application companies can benefit from lower input costs.

4. Boosting reputation, talent and capital

- In a world where you could fit all of the elite AI researchers (about 5,000) into a mid-sized concert hall, and where companies are racing to attract capital to fund exponentially increasing compute costs, a leading open model demonstrates technical competence, attracts researchers and polishes the case for fundraising or an initial public offering.

5. Increasing geopolitical influence

- Open models can spread a country's technical standards, software ecosystem and cultural standards internationally. For China, capable open models reduce its dependence on US APIs and present it as the provider of choice to emerging markets. France has championed Mistral's open models as a European alternative. The US is not sitting idle either. OpenAI itself released smaller, open-weight gpt-oss models as a defensive move last August, and Thinking Machines launched Inkling, its own open model, last week.

5. What comes next?

History shows that formats can coexist where each offers something different.

- The contest between doesn't have to be an either/or. With so much at stake, OpenAI's honeymoon of a near-monopoly on generative AI in the 18 months or so after the launch of ChatGPT in November 2022 was never going to last.
- This is in line with the oft-quoted Jevons Paradox, where more efficient steam engines increased, not decreased, demand for steam power – and hence coal. Similarly, demand for compute will likely continue to surge as the market bifurcates into a handful of cutting-edge proprietary models on the one hand and a myriad of smaller open-source models on the other – just like trains and cars both found their market and continue to live side by side. No one ever said: "That's enough intelligence for now".

But both will exert a positive gravitational force on each other.

- Open models will force proprietary models to keep a lid on prices and remain flexible; proprietary models will set the benchmark for industrial-grade quality, safety and fully integrated products. The competitive moat will increasingly move away from raw weights and towards the balance of proprietary data, product design, memory, security and efficiency.

The greatest unknown in this uneasy coexistence is the direction of AI regulation.

- A growing backlash against AI in the US and the West significantly raises the chance of dramatic policy that could significantly affect its trajectory. While a handful of key US model makers presents a large target for legislators, they are also able to speak with one voice in a way that is not possible for fragmented, often foreign, open-source providers.
- For example, Anthropic released an Advanced AI Framework last month that called for obligations on frontier AI developers to test their models for catastrophic risks, engage external evaluators and disclose incidents on an ongoing basis. Their message echoes warnings from scientists and economists ranging from job losses to extinction.
- But critics say onerous regulations could favour incumbents with deep enough pockets to cover the costs and would stifle innovation, putting national security at risk. Former White House AI tsar David Sacks posted over the weekend on social media that there was "no reason to limit American models on tasks that Chinese models handle without issue. We're only making ourselves less competitive".

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Appendix 1

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El Nino: The next supply shock?

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Authors

Henry Allen
Macro Strategist

Current forecasts point to one of the strongest El Nino events on record in Q4. That matters because El Nino events can act as a multi-dimensional supply shock, with a causal link to higher food and energy prices, alongside broader supply-chain disruption. This would be problematic at any point in time, but it's a particular issue today because of existing supply-chain stresses and the backdrop of above-target inflation. So a very strong El Nino event would be another negative supply shock, at a point when the global economy has limited room to absorb one.

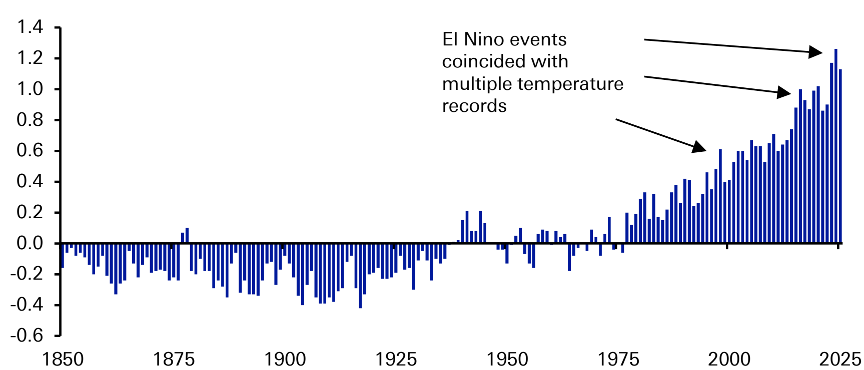
This paper explores what an El Nino is, and why a very strong one later this year would be so concerning. It also takes a look at what happened on previous occasions, and the channels through which it affects the broader economy and financial markets.

We also consider the potential consequences this time round. Much will be weather dependent, but the precedents from the 1970s show how an El Nino event can coalesce with other shocks to keep inflation persistently above target, just as we've seen in recent years. Moreover, this is particularly damaging for many EM economies, where food can make up as much as a third of consumer expenditure.

What is an El Nino?

- An El Nino event is the warm phase of the El Nino-Southern Oscillation (ENSO), which is a variation in winds and sea surface temperatures in the Pacific Ocean. This oscillates between warming phases (called El Nino) and cooling phases (called La Nina).
- The warming El Nino phase occurs when there are unusually warm sea surface temperatures in the eastern Pacific. These warmer waters mean that the Pacific jet stream moves south relative to normal, which creates changes in weather patterns and ecosystems.
- We can see the effect on global temperatures too. Even as temperatures have ben rising anyway over recent decades, the record years have often coincided with El Nino events, including 1998, 2016, and 2024.
- Unfortunately, El Nino events are correlated with a higher frequency of natural disasters, with more people affected by disasters in El Nino years relative to neutral years. Flooding is one that sees a particular increase, but it is also associated with temperature changes too.
- The effects of an El Nino vary considerably by region. In the US, it normally coincides with wetter conditions in the southern US along the Gulf Coast. In stronger episodes, this can extend to California and the Southwest too. But in other regions it is more likely to cause drought, like Australia, whilst India sees weaker monsoon rains.
- In practical terms the main effect of an El Nino is disruption to food production. But there are other effects too. El Ninos have been correlated with the spread of climate-sensitive diseases, and one study found that the El Nino in 2015 contributed to the spread of the Zika virus since it created conditions beneficial for mosquito-borne transmission.[1] Furthermore, natural disasters like flooding increase the frequency of poor hygiene conditions or undernutrition, so there is an impact there too, and El Ninos have been linked to many famines and even conflicts.
- For markets, an El Nino is capable of hitting agricultural production, hydropower, shipping, and even has effects on public health and political stability. In other words, it operates as a multi-channel supply shock.

Figure 1: Global land and ocean average temperature departure with respect to 1901-2000 average (degrees celsius)



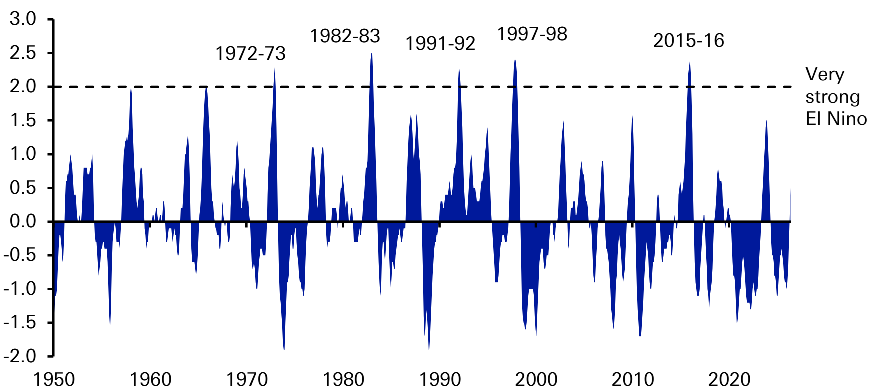
Source : National Centers for Environmental Information, Deutsche Bank

1 Caminade C, et al. Global risk model for vector-borne transmission of Zika virus reveals the role of El Niño 2015. Proc Natl Acad Sci U S A. 2017 Jan 3;114(1):119-124.

What's particularly bad about this one?

El Nino cycles typically repeat every few years with irregular frequency. However, today's episode is particularly concerning because forecasts suggest it will be a very strong one, ranking among the most serious in recent history.

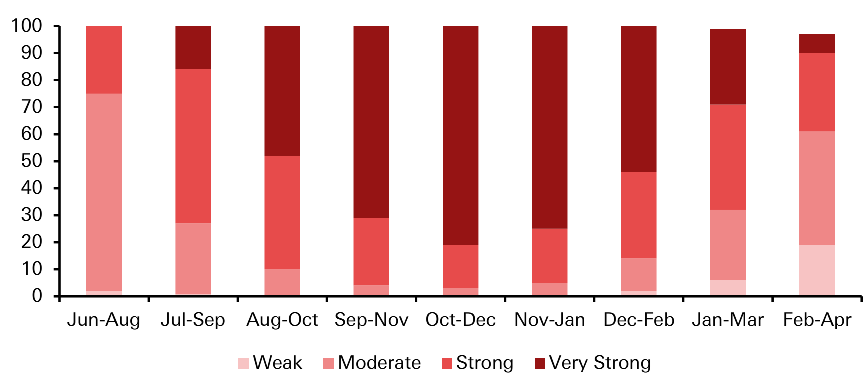
Figure 2: Relative Oceanic Niño Index



Source : Climate Prediction Center, Deutsche Bank

Among others, the US Climate Prediction Center have said this could be one of the strongest in over 75 years, with an 81% chance of it reaching the "very strong" threshold over Oct-Dec. It even has a 73% chance of reaching the "strong" threshold of 1.5C as soon as the July-September period.

Figure 3: US Climate Prediction Center forecasts (%) for El Nino



Source : Climate Prediction Center, Deutsche Bank

A very strong El Nino means it's more likely to shift rainfall and temperature patterns across multiple regions simultaneously. And from a macro perspective, the risks are non-linear. For instance, a modest food shock can be absorbed via inventories, imports or substitution. But a large shock propagating across multiple countries can't be absorbed as easily, with second-round inflation effects more likely to result.

The timing is particularly bad right now, because the global economy is already facing significant supply-chain stress. Most notably, the closure of the Strait of Hormuz has led to a spike in oil and gas prices this year. And even if the El Nino is temporary, the resulting inflation shock will push inflation higher at a time when it's already above target in many countries.

These risks are already being considered by policymakers in decision making, particularly in emerging markets. Indeed, this year has seen central banks in India, Peru and Colombia all reference El Nino in official publications.

What happened on previous occasions?

2023-24: Supply-chain disruption and record temperatures occurred with the last El Nino.

- The most recent El Nino offered a clear example of what can happen, and this didn't even hit the "very strong" threshold.
- 2024 was the warmest year on record, and the episode saw severe droughts in parts of the world, including parts of South America, along with severe flooding in East Africa and South America, consistent with the typical redistribution of rainfall during El Nino.
- Meanwhile, the Panama Canal had its 3rd driest year on record in 2023, meaning that authorities had to cut shipping passing through. The disruption to commercial shipping demonstrated how El Nino is more than just a food price story.

2014-16: The last "very strong" El Nino had broader food security and public health implications

- Just as in 2024, the El Nino of 2016 also saw the biggest upward global temperature anomaly to date.
- Ethiopia suffered its worst drought in decades, leaving millions of people in need of food assistance. Moreover, the Pacific hurricane seasons saw 16 hurricanes in each of 2014 and 2015, the joint-highest number of record.
- One study found that El Nino was a contributing factor to the spread of the Zika virus, since it created conditions beneficial for mosquito-borne transmission.

1997-98: The "very strong" El Nino also had major climactic and public health implications

- 1998 was another record year for global temperatures, and included many extremes around the world. For instance, San Francisco saw its wettest rainfall season since 1889-90, whilst Peru saw 16 times its average rainfall.
- Meanwhile, after major flooding in east Africa, there was an outbreak of Rift Valley fever in Kenya, Somalia and Tanzania. The outbreak also led to major losses of livestock too.

1982-83: The El Nino occurs when high inflation is still lingering

- By 1982-83, inflation had fallen substantially from its earlier heights, aided by monetary tightening like that seen under Paul Volcker at the Fed. However, the El Nino didn't help matters, as it caused major climactic disruption once again, including flooding in the United States, and droughts in Indonesia and Australia.

1972-73: The El Nino played a role in raising inflation before the first oil shock

- Although the oil shock of 1973 was the primary factor that drove the stagflation of the 1970s, several other shocks including the El Nino had already occurred beforehand. This was significant because collectively, they served to push up inflation and inflation expectations. For instance, US fiscal spending had already risen in the late-1960s, thanks to the Johnson Administration's Great Society programmes and the Vietnam War. Then in 1971, President Nixon ended the dollar's link to gold, ending the system of fixed exchange rates that had prevailed after WWII, and in February 1973 there was a dollar devaluation. Amidst all these inflationary pressures, the El Nino event further drove up food prices.

Even further back

- Back in the mid-17th century, there was a period when El Nino events happened about twice as often on average. This was an unusually turbulent period of global history that's been termed the "General Crisis". Some historians have proposed that climactic changes such as El Nino played a role in fomenting instability.

How does an El Nino affect the economy and global markets: The main transmission channels

A useful way to think about El Nino is a set of different supply shocks via multiple channels: food prices; energy prices; supply chains; EM balance sheets; and growing risk premia resulting from the heightened risk of social/political unrest. For individual economies, the impact depends on their exposure to food in the consumer basket, and the extent of intervention from policymakers, be that via food subsidies or tighter monetary policy to counteract inflation.

1. Food: The clearest inflation channel

- **Direct supply hit**: El Nino shifts rainfall patterns, raising the odds of drought in parts of Asia and Australia, and flooding in parts of the Americas. This causes damage to crops, kills livestock, and leads to shipping disruption.
- **Emerging market exposure**: The impact of a food shock varies considerably across different economies. In the US for instance, food-at-home makes up just 8% of the consumer basket for CPI inflation. By contrast, in several EM economies, over a third of household expenditure is on food, and an IMF paper[2] found that the inflation response to El Nino was biggest in economies where food made up a higher share of the consumer basket. So emerging markets like India and Indonesia are among those more severely affected. Moreover, food inflation is a problematic category because it's a necessity, and consumers can't easily substitute away from it, as they can with luxury goods.
- **Second-round effects**: Beyond the immediate impact, El Nino also has second round effects on inflation. For instance, governments may cushion the impact of higher food prices with subsidies. That might reduce the domestic hit temporarily, but by further boosting demand and raising fiscal deficits, it pushes the adjustment elsewhere and risks creating further inflation overall.

2. Energy and Hydropower: The less-obvious inflation channel

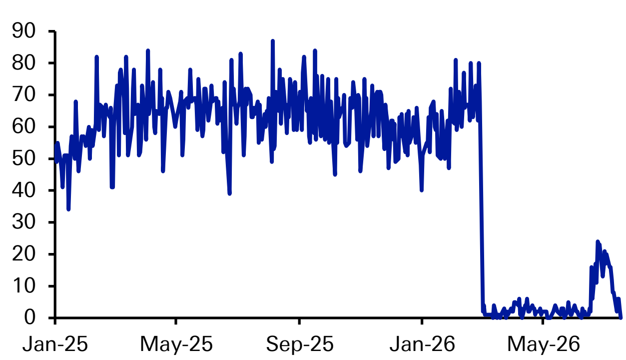
2 IMF Working Paper: Fair Weather or Foul? The Macroeconomic Effects of El Niño (2015), Cashin, Mohaddes and Raissi.

- **Hydropower shortfall**: Lower rainfall in some regions means there's less electricity generated from hydro-electric dams. Moreover, the droughts mean this cut to electricity generation is happening just as heat is raising demand for cooling, and more water is required for irrigation. So there's less supply of energy, just as demand goes up.
- **Fuel-switching channel**: With hydropower output down, this then forces a substitution into other forms of energy, such as gas, coal or oil-fired generation, again raising demand for those other fuels and lifting power prices.

3. Shipping/supply chains

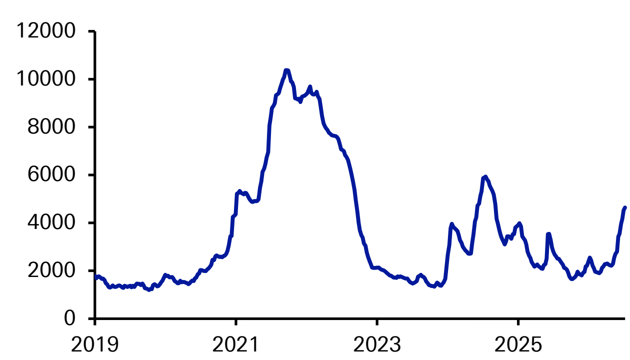
- **Climate effects on supply chains**: The last El Nino in 2023-24 showed how it can impact supply chains, quite aside from a direct impact on the food supply. For instance, Panama had its third-driest year on record in 2023, with rainfall around 30% below average, forcing authorities to restrict the number of ships passing through. This is a point covering around 5% of global maritime trade.
- **Interaction with other chokepoints**: Any supply-chain issues are particularly troublesome today, because the Strait of Hormuz has seen traffic almost completely stop thanks to the US-Iran conflict. So the more transit points that are impacted, the more difficult it is for other routes to absorb any disruption. Indeed, some measures of shipping rates have already doubled since the start of the year
- **Macro Transmission**: If key routes are restricted or closed, then that raises shipping rates further, increases delivery times, and means firms want to hold more inventories against future disruption. So it's another negative supply shock, on top of that seen from higher food and energy prices.

Figure 4: Strait of Hormuz tanker vessel crossings



Source : Bloomberg Finance LP, Deutsche Bank

Figure 5: World Container Index Composite Container Freight Benchmark Rate (USD/40 Foot Box) Drewry



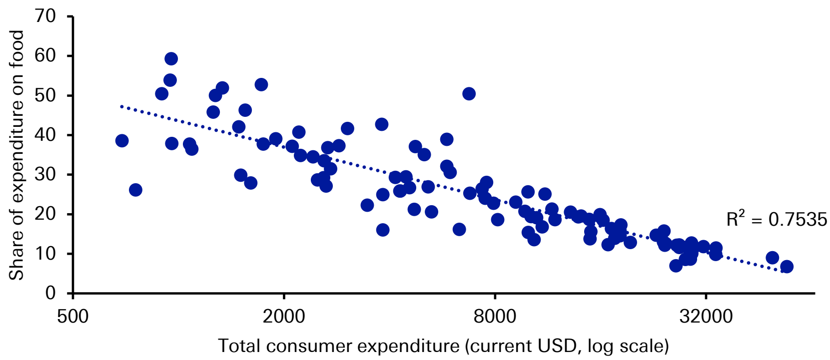
Source : Bloomberg Finance LP, Deutsche Bank

4. EM vulnerability

- **Climate impact**: By virtue of their geographic position, emerging markets are more likely to face the impact of flooding or drought associated with El Nino.
- **Household exposure**: Emerging market economies have more exposure to food prices, so any price shock hits them harder, with food making up over a third of consumer expenditure in several countries.

- **Policy challenges**: This vulnerability makes the policy response to an El Nino event particularly acute. Central banks might raise rates to keep inflation expectations in check and defend the currency, but that doesn't solve the actual supply shock itself. Moreover, if fiscal policy is used to cushion the impact, the shock is then shifted onto public balance sheets, and sovereign risk can become an issue.

Figure 6: Total consumer expenditure vs share of expenditure spent on food, 2023

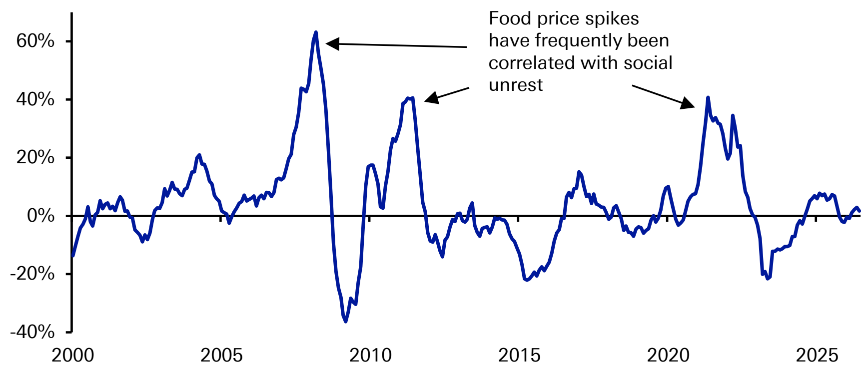


Source : USDA Economic Research Service, Our World in Data, Deutsche Bank

5. Social/political unrest

- **Higher food prices raise the risk of unrest**: Throughout history, higher food prices have often been correlated with political unrest. Clearly it takes many factors to spark unrest, but the squeeze on disposable incomes means that any existing dissatisfaction is magnified. If an El Nino leads to higher food prices, that in turn makes unrest more likely.
- **Recent precedents**: In 2010-11, higher food prices were regarded as a contributing factor behind the Arab Spring. A few years earlier in 2007-08, there was a wave of instability across many EM countries, including Tunisia, Egypt, Cameroon, Ivory Coast, and Haiti. And more recently, the 2022 price shock was a contributor to protests in Ecuador, Sierra Leone, Haiti and Panama.
- **Historical precedents**: Looking further back, several of history's most important revolutions took place against the backdrop of higher food prices. The French Revolution of 1789 came after a succession of harvest failures. The 1848 revolutions in Europe happened at the end of the so-called "Hungry Forties", when there was a severe famine after the failure of potato crops. And in Russia in 1917, food shortages played a key role in fomenting discontent, with supplies hampered in part by the reallocation of farm workers into the military for WWI.

Figure 7: UN Food and Agriculture World Food Price (year-on-year % change)



Source : Bloomberg Finance LP, Deutsche Bank

What does it mean for economies and markets?

A very strong El Nino event would be another negative supply shock when the global economy has limited room to absorb one. The combination of higher food and energy prices will mechanically raise inflation, and broader supply-chain disruption risks creating further price pressures when the Strait of Hormuz has already been blocked.

This is particularly problematic, because experience shows that a prolonged period of above-target inflation raises the risk of inflation expectations becoming unanchored. That in turn can create a self-fulfilling wage-price spiral, as if firms and individuals expect high inflation, then they will set their prices and bargain for wages accordingly.

In several respects, this is reminiscent of the experience in the 1970s. Back then, a series of seemingly unconnected supply shocks (including an El Nino) occurred that collectively kept inflation high. Even though each might have been temporary on paper, implying policy should look through and not over-react, the reality was that the shocks began to feel permanent because they kept on happening. So inflation expectations rose and policy eventually had to react.

Emerging markets are particularly vulnerable as well, with the risk greatest in places where food is a larger part of the consumer basket, and where fiscal space is more limited to accommodate any shock. Moreover, we know from recent history that higher food prices are also linked to higher instances of social unrest, creating further economic uncertainty.

Finally, there is a clear human cost to severe El Nino events. The frequency of natural disasters goes up and there is a greater risk of famine. Moreover, public health issues are often correlated with El Nino events, with several channels in operation, including the heightened risk of cholera after flooding, or the added risk of malaria from conditions more favourable for mosquitos.

Given the forecasts for the strength of this El Nino, this will be an important event to look out for over the months ahead.

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